

Relational attention for a structured semantic latent packet: toward morphology-general robot control

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251 unique simulated embodiments

Counted by `make_figures.py` embodiments: distinct body keys built into a simulator model and simulated in at least one recorded run (validation, teacher screen, collection, training or evaluation). 150 single arms (arm $\times$ gripper; 108 procedural), 19 dual-arm rigs, 80 legged and humanoid bodies (64 procedural humanoids), the Ψ_0 G1 and the ComputerWorld pointer (a UI agent, not a robot); 150 of them are in a training set. All bodies are simulated; none is a physical robot. Breakdown: Table 8.

Abstract

We study robot control in which a planner (System 1) samples a continuous latent packet $z \in \mathbb{R}^{K \times M \times D}$, one slot per knot time and per body assembly, by rectified-flow matching, and a fast body-side controller (System 0) realizes the same packet as native joint targets from proprioception and touch alone. The packet is supervised to carry task meaning (target, contact, held object, goal) through probes. Every attention layer adds declared relational structure to its logits: the logit is $\langle q_i, k_j \rangle / \sqrt{d}$ plus a sum of factor terms (PaPE-style relative geometry, kinematic graph, contact, support and force flow, grasp, task-gated next contact, UI structure). The terms come from one declarative registry of 76 factors (75 implemented) that also drives relation-aware data generation and probe readouts, with a deploy guard that keeps simulator truth out of run-time inputs. Across arms, quadrupeds, hexapods, humanoids and a 3D UI world we record that semantic supervision raises arm competence (+0.23 [0.18, 0.28], D-134), carries a task-context halt request into legged behaviour (D-124) and lets packet edits partially steer a pointer (D-142), but that the latent route does not beat behaviour cloning when adapting to a new arm (D-135/D-136). A pre-registered campaign (humanoids, arm diversity, Ψ_0 , pointer, relation factors) is running. Its recorded outcomes so far are mostly negative or inconclusive: the arm-diversity lineage gate failed, the structured Ψ_0 head reaches 7/20 against 19/20 for direct fine-tuning, a relation-factor comparison was halted at a floor, and no humanoid transfer or relation-factor result exists yet; every in-progress node is labelled as such.

Status, 4 October 2026. This is a living report. The training campaign of Sec. 7 is ongoing; results are recorded to the snapshot of 4 Oct 2026 (campaign `STATUS.md` and decision ledger) and later versions will add to them. Anything marked *in progress* is not a result.

★ Contributions.

① **Packet interface.** System $1 \rightarrow z[K \times M \times D] \rightarrow$ System 0, with per-assembly queries computed from public morphology (no per-ID tables); the same tensor is probed, transmitted and consumed.

② **Relational attention as a sum of declared terms.** One registry (field $\times$ operator $\times$ form $\times$ source) drives biases, kernel-compatible augmentations, gates and masks, and a deploy guard.

③ **Relation-aware data.** One privileged state interface, label and scene-part registries, factor-agnostic transforms and a progressive-composition scheduler generate the data each relation needs.

④ **A ledger.** Recorded results with their intervals, including negative ones, and a pre-registered campaign with its gates.

1 Introduction

A robot policy that is meant to move between bodies has to say what it intends in a form that survives the change of body. We take the *latent packet* as that form. A planner emits z ; a body-side controller turns the received z into joint targets using only what the body itself senses. Task meaning therefore has to live on z , and we require evidence for that from probes *and* from causal edits of the received packet.

The model is organized by a naming that follows Ψ_0 [1]: System 2 is an optional frozen vision-language model, System 1 the flow-matching planner, System 0 the packet realizer. Unlike Ψ_0 's System 0 (an off-the-shelf RL tracker), ours is a learned realizer; an RL tracker sits underneath it for legged bodies. The second ingredient is a view of attention as a sum of named relational terms. Relative geometry follows the kernel-compatible position terms of PaPE [2]; graph, contact and task terms follow the relational-inductive-bias view [5]. What is new here is bookkeeping: each term is one registry entry that also says which labels train it, which scene parts generate its data and whether it is a bias, an augmentation, a gate, a mask or only a readout.

What this report claims. (i) The design, with every factor listed (Table 2, Appendix A). (ii) Recorded results from the ledger, each with its decision number and interval (Ta-

Table 1: Packet shape per family ($z[K \times M \times D]$), from the code. Pointer $D=26$ with the v2 key code (T8).

family	K	knot times / horizon	M	D
arm / dual	4	0.1–0.7 s	1 / 2	64
legged / humanoid	4	0.1–0.7 s (0.8 s)	≤ 11	32
Ψ_0 (G1)	5	chunk steps 5–29 (0.6 s)	6	64
pointer	4	–	1	16 (26)

ble 4). (iii) A pre-registered campaign, described as a plan (Table 5). *Not claimed*: that relation factors help (the experiment that tests this is running), that the latent route transfers across bodies (it did not on a new arm), or any physical-robot result (simulation only).

2 Architecture

System 1. An encoder–decoder transformer trained by rectified flow [3, 4]. The encoder reads clean context tokens once per observation. Tokens come in typed banks: *morph* (one per action node, passive joint and assembly), *scene* (tracked objects with uncertainty and an explicit null token), *task* (events, role bindings, predicates, status) and *interact* (touch and receipts). Legged and humanoid bodies use one global token plus per-joint, per-limb, per-foot and terrain-cell tokens; the pointer world uses UI-widget and instruction-character tokens. Only the Ψ_0 family and an optional arm variant read an image, through System 2 (a frozen VLM and a small resampler giving 16 image tokens).

The decoder’s queries are the packet slots (k, m) for knot k and assembly m ,

$$x_{k,m} = z_\tau[k, m] \text{ in-proj} \\ + \text{node}(a_m) + \text{node_from_ctx}(c_m) + h_k, \quad (1)$$

where a_m are the assembly’s public morphology features, c_m its encoded morph token and h_k a knot embedding. The identity of a query is thus computed from public morphology, not looked up in a per-ID table, which is what lets a new body produce new queries. Each block applies self-attention over time and across assemblies, cross-attention to the cached context and an MLP, conditioned on flow time τ . Sampling integrates $\tau : 0 \rightarrow 1$ by 8 Euler steps over *all* $K \times M$ tokens in parallel; the loss is $\|v - (z - \epsilon)\|^2$ with $z_\tau = (1 - \tau)\epsilon + \tau z$. A training-only target encoder E turns demonstrated future actions into the target packet; probes read meaning out of z and supervise it.

System 0. Each control tick, one query per actuated node is built from public morphology and the *current* measured state ($q, \dot{q}$; legs add IMU, own-foot touch and an oscillator). The node queries cross-attend to the packet’s knot tokens, whose times are shifted by the elapsed phase, through the routing mask **route.own_assembly** (a node reads only its own assembly’s knots; legged bodies add the body assembly’s); then self-attention among nodes and an MLP give one normalized joint target per node. System 0 receives *no* task, instruction, scene, image or System 1 state: task meaning arrives only through z , sensor feedback every tick. Clocks: the arm replans every 0.4 s and runs System 0 at 20 Hz over

500 Hz physics; legged and humanoid bodies replan every 0.4 s with a 0.8 s packet at 50 Hz.

Deploy boundary. Every token set separates **fields** (public or estimated) from **labels** (simulator truth). Each field and factor carries a provenance, *public*, *estimated* (a public estimator or the factor’s own probe) or *privileged*, and a deployable policy refuses privileged sources (**assert_deployable**). Ground truth trains the probes; at run time only public and estimated values steer attention.

3 Relational attention bias

Every attention call in the model uses one shared multi-head attention whose logit is

$$A_{ij} = \frac{q_i^\top k_j}{\sqrt{d}} + \sum_{f \in \mathcal{B}} w_{f,h} \gamma_f v_f(i, j) \\ + \langle \phi_q(i), \phi_k(j) \rangle \quad (-\infty \text{ if masked}), \quad (2)$$

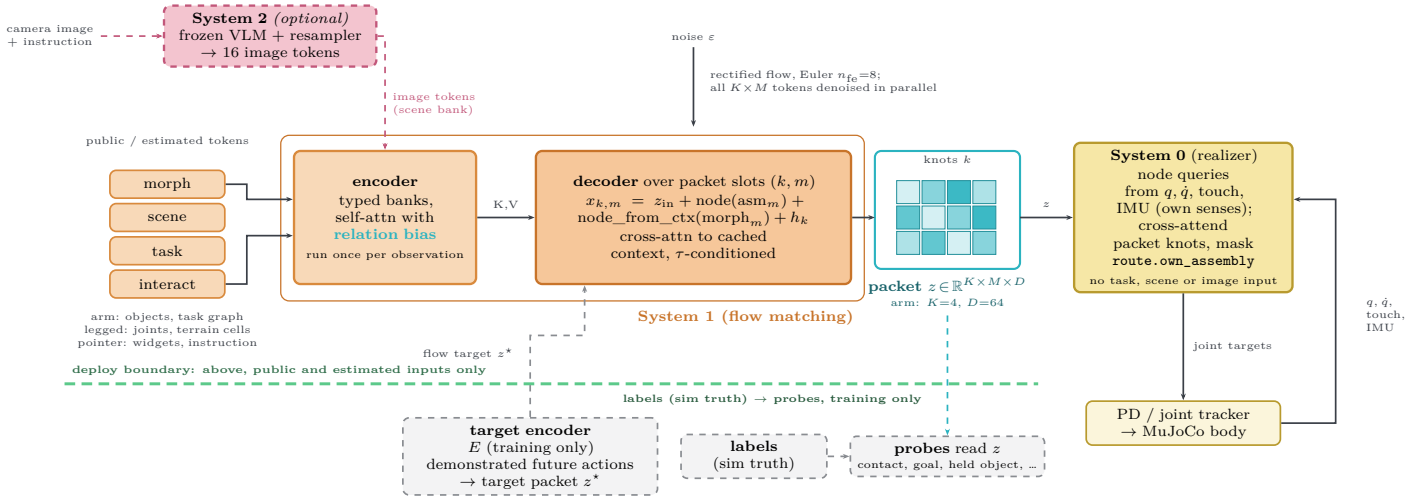
where each registered factor f contributes one term $v_f(i, j)$ in one *form*: an additive *bias* ($f \in \mathcal{B}$), a kernel-compatible *augmentation* appended to q, k (the $\langle \phi_q, \phi_k \rangle$ term, so fused attention kernels still apply), a *gate* $\gamma_f = 2\sigma(u \cdot c + b)$ that scales another term by a task, goal or embodiment summary c , or a hard *mask*. Three non-attention forms (message, embed, readout) reuse the same entries. Every learned coefficient is zero-initialized, so switching a factor on in a trained model is a no-op at step 0.

Operators. A factor is an entry over about a dozen operators, which are the only code per relation *kind*: **diff**, **sqdiff+diff** (PaPE), **rel_rot**, **align**, **order**, **same**, **sim**, **edge**, **hop**, **ancestor** (closure), **flow**, **bilinear** and **unary**. The PaPE term in its compact kernel-compatible form is

$$v_{\text{PaPE}}(i, j) = -(r_j - r_i)^\top M_i (r_j - r_i) + \tilde{b}_i^\top (r_j - r_i), \quad (3)$$

with $M_i \succeq 0$ diagonal in a learned frame and $\tilde{b}_i$ a learned direction; it splits into a *distance* part and a *direction* part (the first two panels of Fig. 2). Graph terms come from declared morphology (**kin.ancestor**, siblings, same assembly) and from the supplied task graph (actor, patient, destination edges). Pair terms that cannot be read from public inputs (contact, support, held-by, handover, next contact) are **bilinear**: $\langle Ux_i, Vx_j \rangle$ is at once the attention term and the pair probe, so the bias *is* a supervised attention subspace, and its estimate can feed a graph factor (**ix.support**→**ix.force_flow**) with source *probe*, deployable. **task.next_contact** is gated by the task so that the same scene shows several candidate edges and the bias narrows to the one the task needs.

Sites and controls. Factors attach at the encoder (**ctx>ctx**), decoder cross- and self-attention (**act>ctx**, **act>act**) and System 0 routing (**node>knot**). Uniform controls apply to any factor: **off**, **zero**, **rewired** (keys permuted, degree preserving), **shuffled**, **reversed**, and for message factors **serialized**, the same facts as hashed text (the equal-information baseline).



clocks. arm: System 1 replans every 0.4 s (4 knots), System 0 at 20 Hz, physics 500 Hz. legged / humanoid: replan every 20 ticks (0.4 s), System 0 at 50 Hz.

Figure 1: System overview. System 1 encodes typed context tokens (public or estimated fields only) once per observation and denoises the packet z in parallel; System 0 realizes the same z from the body’s own senses with no task, scene or image input. Dashed boxes are optional (System 2) or training-only (target encoder, labels, probes: simulator truth, never a deployed input).

Family	Impl.	Data gen	Attn	Readout	Planned
structure	9	0	7	0	4
geometry	5	5	5	4	3
kinematics/body	8	0	8	0	1
interaction	5	5	5	4	6
task/procedure	15	1	15	1	3
locomotion	2	2	1	2	0
UI	5	5	5	1	0
time	0	0	0	0	3
probe readouts	26	0	0	26	0
prop./aff./lang.	0	0	0	0	8
total	75	18	46	38	28

Generated by `make_figures.py` from the registries at `c18101e8`.

Table 2: Relation factors by family. Counts are computed from the registry; a factor can count in several columns. Planned = declared in the registry or catalogued (waves W2, P), not implemented. The full per-factor table is Table 6.

Probes and diagnostics. Probes are readouts in the same registry. Attention maps and probe accuracy are diagnostics: causal use is shown by editing the packet or the context and rolling out (Section 6, D-142, D-124). Fig. 2 shows factor *values* computed from real simulator states, not learned attention; Fig. 5 covers every factor.

4 Relation-aware data and curriculum

Each relation may need data that decouples one invariance, so data generation is as declarative as the factors. A single privileged `StateView` (capability `privileged_truth`, implemented per backend) exposes poses, contacts, joints, cameras and UI trees for *supervision only*. Four registries sit on it: **LABELS** (18 label functions, each with a provenance `gt`, `task` or `estimator`), **PARTS** (7 composable scene components: `table_objects`, `camera_depth`, `grasp_target`, `stack`, `terrain_steps`, `cw_viewport`, `cw_depth`, each declaring the dynamics it activates, requires and conflicts with), **TRANSFORMS** (6 factor-agnostic sample transforms: `reveal`,

`surprise`, `cf_swap`, `noise`, `occlude`, `subsample`) and one composition operator that returns a scene covering a requested set of active dynamics. A factor entry names its `label` and its `gen` parts or transforms; decoupling pairs are copies of a scene that differ only in one factor’s value.

Progressive composition. One scheduler samples the generated rows. Every factor starts at composition depth $k=1$ (isolated parts); it is promoted to $k+1$ when its competence $c_f \geq \theta_f$ for two consecutive intervals; per-factor shares are $\propto s_{\min} + s_f$, clipped to $[2\%, 35\%]$, with struggle $s_f = (1 - c_f) + 0.5 p_f + i_f + e_f$ (plateau, interference and attributed failures); the full-task share follows a rising floor. Steering is a small grammar (boost, pin, freeze, drop, revert) logged for exact replay. *State*: implemented and plumbing-smoked; the campaign run that measures competence by depth and interference (T9) is in progress, so there is no curriculum result in this report. The `cf_swap`, `reveal` and `surprise` transforms produce the decoupling and progressive-information variants of the contact and next-contact labels.

5 Environments, bodies and tasks

All simulation is native MuJoCo [6] with robot models largely from MuJoCo Menagerie [7] (plus a batched Warp legged env and the Ψ_0 SIMPLE benchmark on Isaac Sim) behind one environment, one policy and one task interface; one harness runs any accepted policy $\times$ environment $\times$ task triple and records declined pairs with reasons. Table 3 lists the bodies and tasks and Fig. 3 shows one frame of each environment. Evaluation routes use matched seeds: R0 scripted teacher (privileged), R1 oracle packet E (expert chunk) (diagnostic, not deployable), R2 generated packet (deployable), plus a behaviour-cloning (BC) positive control and task-context edit suites. Splits are sealed before results.

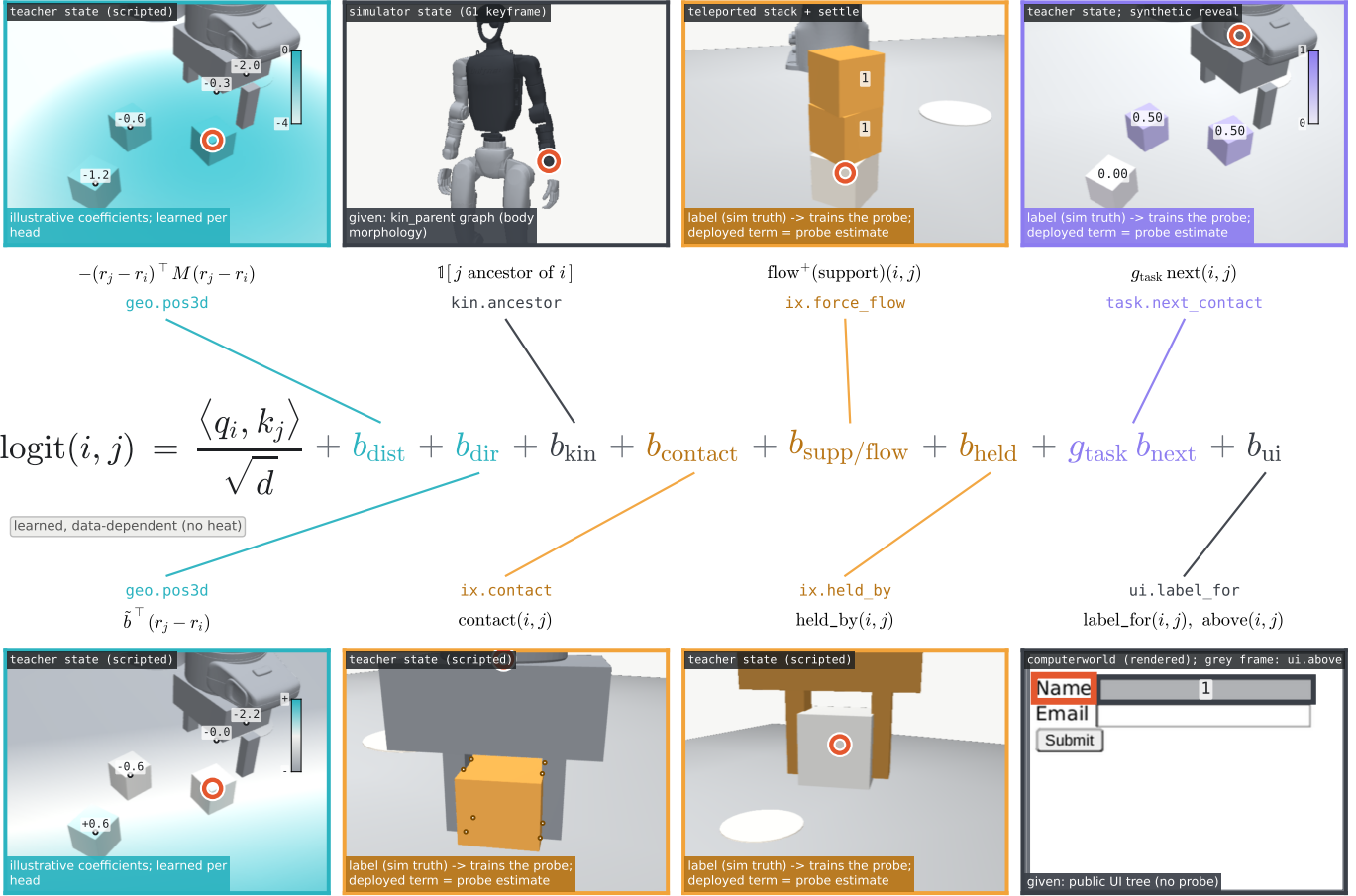


Figure 2: The attention logit, one term at a time. Each panel stretches one term of $\text{logit}(i, j)$ for a single query token (ring) over a real simulator state; frame colour is the relation family (geometry, kinematics/UI, contact/support/grasp, procedure). The content term $\langle q_i, k_j \rangle / \sqrt{d}$ is learned and data-dependent, so it has no panel. *The values shown are computed by the factor operators of `rrp.policies.relations` on real simulator states; they are not trained attention.* Panels 1–2 run the PaPE kernel with hand-set, illustrative coefficients (the model learns them per head); panels 3 and 8 are public, given structure (G1 kinematic tree; computerworld widget tree). For probe-sourced pair terms (panels 4–7) the colour shows the simulator-truth label that trains the probe; the deployed term is the probe’s estimate, which is not shown. Scenes: scripted-teacher Panda pick-and-place states (1, 2, 4, 6, 7), a teleported three-cube stack (5), a G1 keyframe (3), a rendered desktop UI (8). Terms whose form is *aug* (`geo.pos3d`, `ix.contact`, `ix.held_by`, `task.next_contact`) enter the logit as $\langle \phi_q(i), \phi_k(j) \rangle$ in Eq. (2); the panel shows that value. Planned relation factors (e.g. `time.same_track`) are omitted.

6 Results so far

Table 4 lists every result recorded to the snapshot of 4 Oct 2026, with the setting, the comparison and the interval recorded in the cited decision. Results come from three environments (arm pick-place, legged locomotion, ComputerWorld pointer) and from our fine-tunes of one third-party pretrained humanoid policy (Ψ_0); none of them is a positive cross-body transfer claim. Source labels follow the repository convention: *learned latent* and *learned BC* are our policies under test; *scripted teacher* and *privileged oracle* are references; *upstream* marks third-party released weights that we evaluate but did not train.

Semantic packet supervision helps within the training bodies. On four arm bodies and two training seeds the latent route with semfix succeeds in 425/480 R2 episodes against 314/480 for the capacity-matched nosem control, a difference of +0.23 [0.18, 0.28], with semfix ahead in

8/8 body×seed cells (D-134). The latent route stays below the BC control on panda (135/180 against 58/60) and the binding (rebind) effect is seed-dependent for both variants. An earlier “semantic supervision hurts” result was a confound: the unbounded probe loss shrank System-0 updates to ≈ 0.002 – 0.004 against 0.70–0.92 for nosem; with the bounded loss the three variants read 81/90, 83/90 and 21/90 (bounded, nosem, original; D-085, D-087). Comparisons made before that fix are not used here.

The packet is causally used, in two environments. In legged locomotion, adding a “halt” request to System 1’s task context (System 0 has no task input, so the request can reach the body only through z) changes forward progress over 2–5 s by -0.40 m [$-0.50, -0.30$] on anymal_c and -0.66 m [$-0.72, -0.59$] on go2 for semfix relative to nosem, ordered in 3/3 seeds, while an irrelevant context change moves the body by at most 0.02 m (D-124). In ComputerWorld a probe-guided edit reduces the distance to the original target from 163 px to

Table 3: Environments, bodies, tasks, the relation families whose factors resolve in each env’s net family (implemented factors: total, net-side + probe), and the status of each track.

env	bodies	tasks	relation family	status
arm mujoco/arm	panda, sawyer, ur10e, ur5e, vx300s, parm5/6/7 variants (5), 24 procedural pa2s* (pg2 / tf3 grippers); sealed targets panda_tf3, xarm7	pick_place, reach_pose	arm 42 (33+9)	teacher + learned; transfer split primary_v1 / armdiv
dual mujoco/dual	two mounted arms (pairs of the arm pool) or one dual body	support_insert, handover, assign_left, assign_right, pivot_against_surface	dual 43 (34+9)	parked, D-146: eval of external checkpoints; scripted teachers only
quad mujoco/legged	go2, anymal_c, a1, hexapod6, pquad4, sprawl4, sprawl8, hexapod6_long	waypoint_contact, foothold_steps	legged 21 (16+5)	teacher + frozen tracker
humanoid mujoco/legged	t1, g1, h1, op3, apollo, adam_lite, talos + phum procedural	9 h_* tasks (h_steps, h_gap, h_walk, ...); held out: h_steps_carry, h_gap_cart	humanoid 21 (16+5)	teacher + frozen tracker; campaign in progress
tracker warp/legged	same legged and humanoid bodies, GPU-batched	locomotion, h_steps, h_gap	legged / humanoid (as above)	tracker training only
SIMPLE simple	g1_simple (G1, 36-d command)	6 G1Wholebody* tasks	psi0 19 (10+9)	Ψ_0 SIMPLE on Isaac Sim; path-traced renderer caveat; step-1 reproduced on 3 of 6 tasks
CWorld computerworld	cw_pointer (cursor, keys)	calc_sum, open_type, drag_window, fill_form	pointer 10 (7+3)	scripted teacher; procedural strings v3

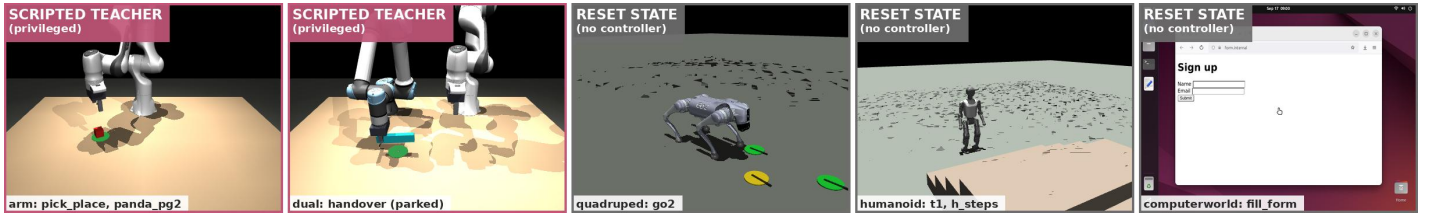


Figure 3: The five environments. One rendered frame each. The arm and dual-arm frames are mid-episode states of the scripted teacher (privileged source); the quadruped, humanoid and computerworld frames are reset states with no controller running, so they show the scene, not a rollout.

41 and 22 px for semfix; a random edit of equal norm gives 102–105 px, and nosem edits behave like random ones (D-142). The edits are partial (the probe reads the new target in 8–10 of 25 episodes) and task success is unchanged. These are rollouts after interventions, not probe accuracies.

Zero-shot transfer to a new arm is not supported.

On sealed xarm7 targets, latent and BC policies both score 0/200 zero-shot. With target demonstrations BC fine-tuning reaches 69–192 of 200 depending on target, whereas flow fine-tuning gives 0 and a System-0 refit at most 38 (D-135). Joint flow and realizer adaptation, added *after* D-135 and therefore post-hoc, beats the refit in 12/12 cells (595 against 130 of 1,200 for semfix) but stays below BC fine-tuning in 12/12 (595 against 842), with every Newcombe upper bound below zero (D-136). On the known arm with a new gripper (panda_tf3) semfix reaches 154/200 zero-shot, nosem 32 and BC 199 (D-135). A dev-seed diagnosis (D-135 addendum) attributes the D-135 failure to the protocol adapting one module at a time and not to an inability of the latent route to adapt; even so, on dev seeds the latent route adapts about half as fast per update as BC and never beats it on a new body in the data so far.

Pointer control: close to BC in-distribution; unseen strings fail with the v1 packet. On the sealed `sealed_id` set ($n=400$, one training seed, sealed once) BC solves 398 (0.995 [0.982, 0.999]), semfix 372 (0.930 [0.901, 0.951]) and nosem 368 (0.920 [0.889, 0.943]); a learned System 1 driving the scripted System 0 (**eng**) solves 185 (D-142). The semfix–nosem difference is not distinguishable ($p=0.68$, McNemar); BC is ahead of semfix ($p=2.2 \times 10^{-7}$). On unseen words and names every learned method scores 0/50 on typing and 0/50 on form filling; unseen calculator pairs reach 45–50/50. With the v1 packet the policies do not copy characters from the instruction. Campaign round 1 (T8, development seeds only, below) changes this picture for the v2 key code.

Structure on a pretrained humanoid policy: fixed, but below direct fine-tuning. On TabletopGraspMP, 20 closed-loop episodes give 20/20 [0.84, 1.00] for the released Ψ_0 checkpoint (upstream weights, not ours), 19/20 [0.76, 0.99] for our direct fine-tune and 0/20 [0.00, 0.16] for our first structured head (D-141). The offline diagnosis traced that 0/20 to an integration fault: System 0 barely read the packet and acted as a state-only policy, driven by a commanded-torso state that never occurs in training. After the fix, the pre-registered packet-use gate passed (gap 0.0782;

D-146, D-147) and the closed-loop campaign run (T7, seed 0) gives structured 7/20 [0.18, 0.57] against direct 19/20 [0.76, 0.99], paired McNemar $p=0.0018$ (13 vs 1 discordant; D-147 T7 addendum). The fix worked only partially; structure does *not* match direct fine-tuning on this task (one seed, one task). Open-loop held-out error favours the structured head and does not predict closed-loop success.

Campaign outcomes recorded so far (D-147 addenda). *Arm diversity (T6):* the G3 lineage gate failed. On the four original source bodies (development seeds, pooled over two seeds) the latent semfix route trained on 65 arm keys (v8div) succeeds in 362/480 (0.754 [0.714, 0.791]) against 425/480 for v6, below the pre-registered threshold 0.835 (difference $-0.131 [-0.179, -0.083]$); the track is **failed_hypothesis** and, per its signed protocol, no sealed new-arm cell was run, so H1 and H2 are untested. Ψ_0 *Bend-PickMP (T7):* released (upstream, not ours) 20/20, our direct fine-tune 15/20 [0.53, 0.89], packet-use gate passed (gap 0.1483); the structured closed-loop evaluation is *in progress*. *Pointer round 1 (T8, development seeds, three training seeds, nosem):* with the v2 discrete key code, a learned System 1 with the free key head driving the scripted System 0 types strings from the held-out bucket (open 50/50 on every seed, fill 50, 49, 50 of 50), so D-142’s 0/50 does not reappear on development seeds; the pre-registered copy head is *worse* on drag (mean 5.3 vs 37.3 of 50, 3/3 seeds). Nothing sealed was run. *Relation factors (T9):* the first comparison (geo vs base on the v3dart F0 route) was halted at a floor, 0/360 vs 0/360, which is uninformative about factors; the re-planned comparison on the v6 lineage is *in progress*. *Humanoids (T1–T5):* terrain trackers failed round 2, so **h_steps** and **h_gap** are excluded for this campaign; the scripted-teacher quality gate passed for **h_walk** (0.95) and **h_turn** (1.00) only; on the sealed bodies the scripted teacher over the shared tracker failed its pre-registered check (n1 0/40, toddlerbot_2xc 0/40) and toddlerbot_2xc also failed after Level-1 adaptation (0/20 walk, 0/20 turn), so no sealed humanoid cell has run. The **preset:legged** vs **legged-none** comparison (T5) is *in progress* and carries a fair-input caveat: the legged-none arm never receives the public terrain scan or limb and foot tokens, so a matched-input control (**legged-tokens**: same inputs, every relation term off) was pre-registered and is *running*; no legged-vs-none claim is made before it lands.

7 Pre-registered campaign

The remaining claims are decided by a ten-node campaign (T0–T9; D-147, **research/readiness.md** section B) whose tests and reading rules were written before the corresponding runs. Table 5 gives each node’s state at the snapshot of 4 Oct 2026. GPU work is admitted by a resource broker (at most two concurrent GPU jobs per machine, humanoid first); a failed gate is recorded and not retried with other settings. Sealed splits are hash-pinned and each cell runs once; infeasible or failed cells are reported as such.

Humanoid (T0–T5). Question: with pinned, gated trackers, does the legged relation preset change dev trans-

fer relative to the same recipe without it (**preset:legged** against **legged-none**, at least two seeds, matched data and updates)? A tracker is installed only if it passes the D-112 gate, or the labelled D-147 exception (joint margin ≥ -0.06 , peak force ≤ 4.2 body weights, cost of transport ≤ 2.5 , task evaluation ≥ 0.9). One bounded fix per failed gate is allowed. Terrain-scan trackers failed rounds 1 and 2 (T1), so terrain tasks are closed for this campaign; a task with teacher success below 0.8 is held back (T4), which leaves **h_walk** and **h_turn**. The test is the dev transfer tables (T5, plus a leave-one-out comparison among t1, g1, h1) followed by a written decision before any sealed cell; no humanoid transfer claim is made until then. Because **legged-none** lacks the terrain scan and limb and foot tokens, a matched-input control (**legged-tokens**) is pre-registered; “relation factors help” needs **legged** > **legged-tokens**.

Arm diversity (T6). H1: a latent route trained on 65 arm $\times$ gripper keys (13 v6 keys, 24 procedural arms and two more menagerie arms, each with two grippers) transfers to new arms better than the v6 13-body lineages, zero-shot and at 5, 20 and 100 target demonstrations, and closes the gap to BC fine-tuning. H2: kinematic static features (**kinfeat**) help further. Success is $k/100$ per cell with Wilson intervals; three Newcombe contrasts: (a) v8div – v6 latent per primary target, (b) v8div latent – v8div BC fine-tuning at equal budget, (c) kinfeat – no-kinfeat. A contrast counts only if its interval excludes 0; tiers are not pooled; zero-shot is reported apart from adaptation. Gate G3 requires v8div semfix $\geq$ v6 semfix -0.05 on the original bodies and a BC expert ≥ 0.85 on the new arms. The BC half passed; the lineage half **failed** (362/480 vs 425/480, Sec. 6), so the track is recorded as **failed_hypothesis** and stopped without running any sealed cell; H1 and H2 remain untested.

Packet use in Ψ_0 (T7). Question: does a structured head that reads the packet change closed-loop pick-place relative to direct fine-tuning? The gate requires $\text{err } R(\bar{z}) - \text{err } R(E(a)) \geq 0.05$; a failure ends the node as **failed_hypothesis**. It passed on TabletopGraspMP (0.0782) and BendPickMP (0.1483). The closed-loop reading rule, fixed before the BendPickMP and HandoverTeleop results, is per task: structured vs direct with Wilson intervals and paired McNemar; “structure helps” only if structured > direct with $p < 0.05$, and “no worse” is never claimed from overlapping intervals. Tabletop: structured below direct (7/20 vs 19/20). BendPickMP structured and HandoverTeleop are *in progress*.

ComputerWorld pointer (T8). Dev seeds, 50 per task, three training seeds; a difference counts only if it has the same sign in at least two of three paired seeds and exceeds the larger within-arm seed range. P-SEEDS: is the D-142 ordering stable? P-UI: do **ui.*** factors help (mean gain ≥ 4 points on ≥ 2 tasks, no task losing more than the seed range)? P-COPY: does a copy head fix unseen typing ($\geq 10/50$ on both **open_type** and **fill_form** of **sealed_heldout**)? P-KEY: does a discrete key code help (dev gain above the seed

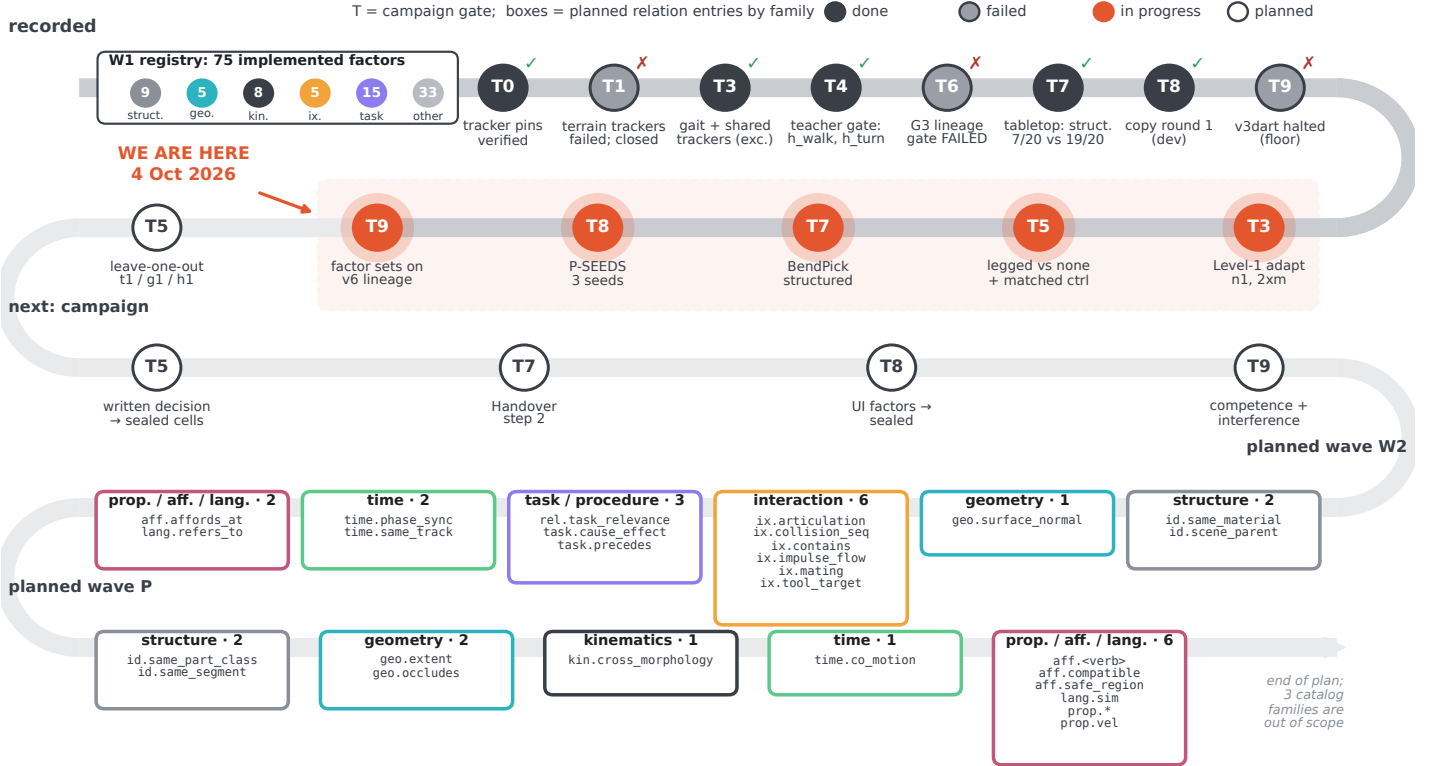


Figure 4: Roadmap as of 4 Oct 2026. The path runs from recorded outcomes (top) to the end of the current plan (bottom). Row 1: the 75 implemented relation factors by family and the recorded gate outcomes, including the failed ones (T1 terrain trackers, T6 G3 lineage gate, T9 v3dart run halted at a floor). Row 2: the current front (highlighted; T3 Level-1 adaptation, T5 legged vs none with the matched-input control, T7 BendPick structured, T8 P-SEEDS, T9 factor sets on the v6 lineage), then the leave-one-out comparison. Row 3: the remaining campaign gates, each ending in sealed or pre-registered readouts (Table 5). Rows 4–5: every planned entry of Table 7, clustered by its family, wave W2 before wave P; they come after T9 because T9 is the first test of whether factor sets help. Colours follow the families of Fig. 2 (geometry cyan, kinematics graphite, interaction amber, procedure violet; structure, time and property/affordance/language get neutral extra hues). Filled = done, grey with a cross = failed (recorded, not retried), highlighted = in progress, hollow = planned. Generated by `make_figures.py roadmap` from the relation registry and catalog.

range)? Sealed sets are evaluated once per arm and seed after the dev table is written. The scripted route cannot copy and is excluded from P-COPY. Round 1 (copy vs free key head, dev) is recorded above; P-SEEDS is *in progress* and P-UI is planned.

Relation factors (T9). Question: do the geometry, interaction and task factor sets change arm success relative to the same pipeline without factors? The first run (v3dart, F0-only route) was halted at a floor (0/360 in both arms; uninformative), and the v8div re-plan became moot when G3 failed. The current pre-registration, written before any run, trains the factor sets {none, geo, ix, task} × seeds 1, 2 on the suffix of the v6 semfix lineage, with **none** on current code as the primary control. A set helps or hurts only if the Newcombe 95% interval of set – control on the 480 pooled primary cells excludes 0, and the result is valid only if every scheduled factor reaches probe competence ≥ 0.5 ; per-factor competence by depth and interference are reported alongside. State: *in progress*. Probes and attention maps are diagnostics; causal use is shown by edits and rollouts (Sec. 6).

8 Limitations

The evidence is narrow. (i) *No cross-body transfer claim.* Zero-shot transfer to new arms is 0/200 for every method, the latent route has not beaten BC on any new body, the arm-diversity lineage gate failed before any sealed new-arm cell, and no sealed humanoid cell has run (the sealed bodies checked so far failed the scripted-teacher check). (ii) *Seeds.* Most ComputerWorld numbers are one training seed (D-142); the legged result has three seeds per body (D-124), with exact seed-ordering $p=0.05$. Intervals cover evaluation episodes, not training-seed variance. (iii) *Post-hoc analysis.* The joint-adaptation method of D-136 was added after the sealed results of D-135 on the same scenes; it is labelled and not a pre-registered test. (iv) *Partial edits.* Packet edits move behaviour part of the way (the probe reads the edit in 8–10 of 25 cases) and are demonstrated on calculator and form tasks only. (v) *Unseen strings.* With the v1 packet all learned pointer methods fail on unseen typing and form text (0/100, sealed). The v2 key code types held-out-bucket strings on development seeds, but no sealed v2 evaluation exists yet. (vi) *Pretrained humanoid.* After the integration fix the structured head reaches 7/20 against 19/20 for direct fine-tuning on one task and one seed; the other Ψ_0 tasks are

in progress. (vii) *Humanoid trackers*. Terrain trackers failed their gates, so terrain tasks are out of this campaign; the installed gait and shared trackers are labelled exceptions to the D-112 gate. The legged vs legged-none comparison mixes relation factors with extra inputs until the matched-input control lands. (viii) *Simulation*. All results are simulated, with scripted or privileged teachers; sim-truth labels train probes and are not deployable inputs. No physical robot was commanded. (ix) *Snapshot*. This is a living report of an ongoing campaign (snapshot 4 Oct 2026); running nodes may finish with results that are not shown here.

References

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- [5] P.W. Battaglia *et al.* Relational inductive biases, deep learning, and graph networks. arXiv:1806.01261, 2018.
- [6] E. Todorov, T. Erez, Y. Tassa. MuJoCo: a physics engine for model-based control. IROS, 2012.
- [7] Google DeepMind. MuJoCo Menagerie: a collection of high-quality simulation models for MuJoCo. github.com/google-deepmind/mujoco_menagerie, 2022.

A Relation tables (generated)

Both tables are written by `make_figures.py table_relations` from the live registries and the candidate catalog; nothing about an implemented factor is typed by hand. Fig. 5 draws every row of Table 6 as one tile: the factor’s term for one query token on a real simulator or environment state, computed by the same operator and label code the model uses, with hand-set coefficients where the model learns them. It extends Fig. 2 from eight panels to all implemented factors and, like it, shows no trained attention. Badges separate public (given) terms from probe-sourced ones, whose tiles show the simulator-truth label that trains the probe (training-only); the deployed term is the estimated-probe value. Under each tile its caption gives what the term encodes on the shown state, the operator and form with the hand-set coefficients, the term’s source and its equation. Tiles marked illustrative are not fully computable from shipped code or data; their caption gives the reason.

B Embodiment count (generated)

Table 8 is written by `make_figures.py embodiments`, which reads the teacher-validation records, the arm admission screens, the dual-arm teacher validation, the legged body catalog, the tracker training metadata (host store and the host copy of the peer archive) and the frozen splits, and keeps the per-class key lists in `figures/embodiments.json`.

Embodiment class	Sim.	Trained	Evidence
single arms: menagerie + fixture arms, each with each gripper module (pg2, tf3)	42	17	teacher validation (20 seeds) + armdiv admission screens
procedural arms <code>arm_gen_v2</code> (<code>pa2s<seed></code> , train and sealed seeds) × gripper	108	48	armdiv admission screens (built keys only)
dual-arm rigs (unordered arm pairs) + ALOHA	19	6	dual teacher validation, assign split
procedural legged (quad / hexapod / octopod)	5	5	legged catalog (physics validated)
menagerie quadrupeds and bipeds	4	2	legged catalog (physics validated)
menagerie humanoids	7	6	legged catalog + tracker training metas
procedural humanoids <code>phum_<seed></code>	64	64	shared-tracker training metas
G1 + Dex3 through the Ψ_0 / SIMPLE interface (<code>g1_simple</code>)	1	1	D-141 closed-loop evals
ComputerWorld pointer (<code>cw_pointer</code>)	1	1	D-142 evals
total	251	150	

Generated by `make_figures.py embodiments` at 908a0430; keys in `figures/embodiments.json`.

Table 8: Embodiments counted for the banner. Sim. = distinct body keys built into a simulator model and simulated in at least one recorded run (validation, teacher screen, collection, training or evaluation); Trained = of those, in a frozen training pool or a tracker’s training set. Arm keys are arm × gripper module; procedural arms are 54 built `arm_gen_v2` seeds; generator seeds whose build failed, dual left/right swaps, and declared but not yet run bodies (humanoid sealed targets, `phum` sealed seeds) are not counted.

Table 4: Results so far (recorded results only; interim numbers of running campaign nodes are not listed). “Latent” = learned System 1 → packet → learned System 0, deployable route (R2); “BC” = learned behaviour-cloning control on the same demonstrations; “teacher”, “oracle” = scripted / privileged references, never policies under test. Intervals are the 95% intervals recorded in the cited decision (Wilson for rates, Newcombe for rate differences); “counts” = no interval recorded. semfix = semantic packet supervision with the bounded probe NLL (D-085); nosem = capacity-matched control without it.

Question	Setting (env, bodies, seeds, n)	Comparison	Result [95% CI as recorded]	Decision
Does semantic packet supervision help the arm latent route?	Arm pick-place, 4 bodies $\times$ 2 training seeds, 240 R2 episodes per lineage (480 pooled per variant); v6 data, teacher v2, grasp v2.1	latent semfix vs latent nosem; BC reference	semfix 425/480 (0.89) vs nosem 314/480 (0.65): +0.23 [0.18, 0.28]; semfix wins 8/8 body $\times$ seed cells. Binding (rebind) is seed-dependent for both. Latent is below BC on panda (semfix 135/180; BC 58/60; BC 60/60 on parm6).	D-134
Does the latent route transfer <i>zero-shot</i> to a new arm?	Sealed targets xarm7_pg2, xarm7_tf3 (new arm), panda_tf3 (new gripper, known arm); 2 seeds $\times$ 100 sealed scenes, k/200 pooled	latent semfix / nosem vs BC; zero-shot, then flow SFT, system-0 refit, BC SFT at 5/20/100 target demos	xarm7: zero-shot 0/200 for latent and BC; BC SFT 69–140 (pg2), 143–192 (tf3); latent refit $\leq$ 38, flow SFT 0. panda_tf3: semfix zero-shot 154, nosem 32, BC 199. counts. Not supported.	D-135
Does joint flow + realizer adaptation close the gap? (<i>post-hoc</i>)	Same sealed xarm7 scenes; 2 seeds; 6 target $\times$ budget cells per variant (1,200 episodes); method added <i>after</i> D-135	joint adaptation (update-matched 150/300/600) vs refit-only vs BC SFT	Beats refit in 12/12 cells (semfix 595 vs 130 of 1,200); below BC SFT in 12/12 cells (semfix 595 vs 842; gap -0.07 to -0.34 ; nosem -0.30 to -0.80 ; every Newcombe upper bound < 0). semfix 595 vs nosem 151. Latent never beats BC.	D-136
Does the packet carry a task-context “halt” into legged behaviour?	mujoco/legged, contact v2, anymal_c and go2, 3 training seeds each; R2 success $n=90$ per variant; forward-progress change over 2–5 s	latent semfix – nosem, pooled; halt request added to the task context	anymal_c -0.40 m [-0.50 , -0.30]; go2 -0.66 [-0.72 , -0.59]; 3/3 seeds ordered (exact $p=0.05$); irrelevant-context control ≤ 0.02 m; success comparable (anymal_c 78 vs 83/90, go2 87 vs 87). t1 under sourced limits: untestable (BC 0/30).	D-124
Was “semantic supervision hurts” real? (confound found and fixed)	t1 humanoid (inflated-torque tracker), R2, 3 training seeds, $n=90$	bounded-NLL sem vs nosem vs original sem	81/90 vs 83/90 vs 21/90. The unbounded probe NLL scaled System-0 updates by ≈ 0.002 – 0.004 (nosem 0.70–0.92); earlier sem-vs-nosem comparisons were confounded. counts.	D-085, D-087
Pointer on ComputerWorld: how does the latent route compare?	4 tasks, sealed_id $n=400$ (100/task), 1 training seed, sealed once; teacher 400/400, oracle→scripted System 0 400/400 (references)	latent semfix vs nosem vs BC vs scripted-System-0 route (eng)	BC 398 (0.995 [0.982, 0.999]); semfix 372 (0.930 [0.901, 0.951]); nosem 368 (0.920 [0.889, 0.943]); eng 185 (0.463). McNemar: BC vs semfix $p=2.2e-7$; semfix vs nosem $p=0.68$.	D-142
...unseen words and names	sealed_heldout $n=150$ (50 calc / type / form), 1 seed, v1 packet	all learned methods	Typing and form-filling 0/50 each (0/100 unseen words and names) for every learned method; unseen calculator pairs 45–50/50. No method copies characters from the instruction.	D-142
...causal use of the packet	Probe-guided packet edit, 25 dev episodes each on calc and form, 1 seed	semfix vs nosem vs random edit of equal norm	Approach to the original target: 163 px (no edit) / 102–105 (random) → 41 / 22 (semfix); nosem edits $\approx$ random (128–141 vs 137–144 px). Edits partial (probe reads the new target in 8–10/25). Task success unchanged.	D-142
Does adding structure to a pretrained humanoid VLA work?	Ψ_0 TabletopGraspMP L0 (SIMPLE / Isaac Sim 5.1, path-traced), 10 eval configs $\times$ 2 repeats, $n=20$ per arm	released Ψ_0 (upstream, not ours) vs our direct fine-tune vs our structured head	Released 20/20 [0.84, 1.00]; direct 19/20 [0.76, 0.99]; structured 0/20 [0.00, 0.16], diagnosed as an integration bug (System 0 ignores the packet).	D-141
...after the fix: packet-use gate	T7 phase 1, held-out episodes; gate = $\text{err}(R(\bar{z})) - \text{err}(R(E(a))) \geq$ margin 0.05	stage A with state dropout and permuted-packet hinge	PASS: gap 0.0782 (0.0843 vs 0.0060).	D-146, D-147
...closed loop after the fix	Same task and 20 configs as D-141, campaign T7, 1 seed	our direct vs our structured (generated packet); released (upstream) reference	Structured 7/20 [0.18, 0.57] vs direct 19/20 [0.76, 0.99]; McNemar $p=0.0018$; released 20/20. Structure below direct.	D-147 (T7)
More training arms: source competence kept? (G3)	Arm pick-place, 4 original source bodies, dev seeds, 2 seeds, 480 pooled	latent semfix v8div (65 arm keys) vs v6; threshold v6 -0.05	362/480 (0.754 [0.714, 0.791]) vs 425/480; -0.131 [-0.179 , -0.083]. FAIL (failed_hypothesis); no sealed cell run.	D-147 (T6)
Pointer v2 key code: copy head vs free head	Dev seeds, 50 per task, 3 training seeds, nosem; learned System 1 → scripted System 0	copy vs free key head (P-COPY rule)	Free head types held-out-bucket strings (open 50/50 $\times$ 3; fill 50, 49, 50); copy head worse on drag (5.3 vs 37.3 of 50, 3/3 seeds). Dev only.	D-147 (T8)

Table 5: Campaign T0–T9 (D-147; gates from `research/readiness.md` section B and the D-147 addenda), state as of 4 Oct 2026. States use the repository’s progress vocabulary only. A recorded gate outcome is listed when one exists; interim counts of nodes that are still `running` are deliberately omitted. “L” = body leg length. Teacher checks use the *scripted teacher* (privileged) over a learned tracker.

	Track and run	Pre-registered gate	State	Recorded gate outcome
T0	humanoid: pinned trackers and warm starts	<code>t1:contact_v2</code> and warm-start checkpoints present with the recorded sha256	<code>verified</code>	Pins match; its D-112 report reproduces D-114 (no new run).
T1	humanoid: terrain-scan trackers, <code>h_steps</code> (t1, g1, h1) and <code>h_gap</code> (t1, h1), rounds 1 and 2	D-112 gate; task eval ≥ 0.9 per height/level; labelled exception only for joint margin, peak force, CoT within caps; one bounded fix per failed gate	<code>failed_hypothesis</code> ; <code>h_steps</code> , <code>h_gap</code> excluded; terrain closed for D-147	Round 1: steps t1 2/20 at 0.10L, g1 0/20, h1 10/20 at 0.10–0.15L; gap t1 0/20, h1 20/20 but stand falls and force above cap. Round 2: no body passes (t1 steps 0/100). None installable.
T2	humanoid: wholebody <code>*_ub</code> gait trackers (t1, g1, h1), <code>steps_ub</code> , <code>h_carry</code> teacher check	Lab gate; teacher check <code>h_carry</code> $\geq 10/12$ both sides, <code>h_steps_carry</code> $\geq 8/10$, else <code>blocked_external</code>	gait trackers <code>completed</code> ; <code>h_carry</code> <code>blocked_external</code>	t1, g1, h1 <code>ub_v1</code> installed under the labelled D-147 exception (joint margin 0.0183 / 0.0191 / 0.0160 against 0.02). <code>steps_ub</code> : h1 installed, t1 and g1 fail. <code>h_carry</code> : no plan on g1 / h1.
T3	humanoid: shared <code>morph_v2_ub</code> tracker; sealed-body teacher check; Level-1 adaptation	Gate on every source body; sealed-body teacher $\geq 16/20$ per (body, task), else <code>blocked_external</code>	installed (owner exception); n1 adaptation <code>running</code> , <code>toddlerbot_2xm</code> queued	Gate failures carried in the label (op3, adam_lite, apollo). Sealed-body check: n1 0/40, <code>toddlerbot_2xm</code> 0/40; 2xc after adaptation 0/20 walk, 0/20 turn: <code>blocked_external</code> .
T4	humanoid: teacher-quality gate, collection	Pooled teacher success ≥ 0.8 per task, else held back	<code>completed</code>	Pass: <code>h_walk</code> 0.95, <code>h_turn</code> 1.00. Held back: <code>h_reach</code> , <code>h_squat_pick</code> , <code>h_place</code> (0.33), <code>h_loco_pick</code> (0.05).
T5	humanoid: <code>preset:legged</code> vs <code>legged-none</code> vs BC, ≥ 2 seeds; leave-one-out t1/g1/h1; matched-input control <code>legged-tokens</code>	Dev tables, then a <i>written decision</i> before any sealed cell; factors help only if legged > legged-tokens	<code>running</code> (fair-input caveat on every table)	None yet.
T6	armdiv: BC 1702, kinfeat BC 1701; G4 signature; v8div lineages $\rightarrow$ G3 $\rightarrow$ sealed	BC expert ≥ 0.85 on new training arms; v8div semfix $\geq$ v6 semfix -0.05 on original bodies; each sealed cell once	<code>failed_hypothesis</code> (stopped)	BC gate PASS (117/120 new arms for both). G3 lineage gate FAIL : 362/480 vs 425/480 (threshold 0.835). No sealed cell run.
T7	Ψ_0 : labels $\rightarrow$ stage A $\rightarrow$ packet-use gate $\rightarrow$ heads $\rightarrow$ closed loop; TabletopGraspMP, BendPickMP, HandoverTeleop	Gate gap ≥ 0.05 , else <code>failed_hypothesis</code> ; closed loop: helps only if structured > direct, McNemar $p < 0.05$	tabletop <code>completed</code> ; bendpick <code>running</code> ; handover <code>planned</code>	Tabletop: gate PASS 0.0782; structured 7/20 vs direct 19/20 ($p=0.0018$); released (upstream) 20/20. BendPick: gate PASS 0.1483; direct 15/20; released (upstream) 20/20.
T8	pointer v2: collection $\rightarrow$ copy / key heads $\rightarrow$ seeds $\rightarrow$ UI factors, dev tables	P-KEY, P-COPY, P-SEEDS, P-UI rules (Sec. 7); sealed only after a written decision	collection, copy round <code>completed</code> ; seeds <code>running</code> ; UI <code>planned</code>	Collection 4 $\times$ 6,000 teacher episodes. Oracle packets $\rightarrow$ scripted System 0: 50/50 per task (privileged reference). Copy head worse than free head on drag (dev).
T9	relation factors: {none, geo, ix, task} $\times$ seeds 1, 2 on the v6 semfix lineage	Newcombe 95% of set – none excludes 0; valid only if every factor’s competence ≥ 0.5	v3dart run halted (<code>failed_hypothesis</code> , floor); v8div moot; v6 run <code>running</code>	v3dart: 0/360 vs 0/360, uninformative.

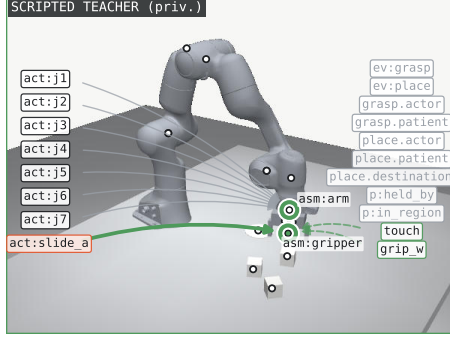
Factor	Operator	Form	Source(s)	Data gen (parts / transforms)	Attn	Read	Nets	Data envs
STRUCTURE / EDGES (9 implemented)								
edge.node_in_assembly	edge	bias	given	N	Y	N	arm,dual,leg,hum	–
edge.same_assembly	edge	bias	given	N	Y	N	leg,hum,psi0	–
edge.same_node	edge	bias	given	N	Y	N	arm,dual,leg,hum,psi0	–
id.same_assembly	same	aug	given	N	Y	N	arm,dual,leg,hum,psi0	–
id.same_body	same	aug	given	N	Y	N	arm,dual	–
id.slot_handle	inert	embed	given	N	N	N	arm,dual	–
msg.incidence	inert	message	given	N	N	N	arm,dual	–
route.assembly_reads	same	mask	given	N	Y	N	arm,dual,leg,hum,psi0	–
route.own_assembly	same	mask	given	N	Y	N	arm,dual,leg,hum,psi0	–
GEOMETRY (5 implemented)								
geo.above	order	bias	given, gt [†]	Y: table_objects	Y	N	arm,dual,ptr	arm,dual
geo.depth3d	sqdiff+diff	aug	probe, given, gt [†]	Y: camera_depth	Y	Y	arm,dual	arm,dual
geo.normal_align	align	aug	given, probe, gt [†]	Y: contact_normal	Y	Y	arm,dual	arm,dual, mj-leg,wp-leg, simple
geo.orient	rel_rot	aug	given, probe, gt [†]	Y: table_objects	Y	Y	arm,dual	arm,dual
geo.pos3d	sqdiff+diff	aug	given, probe, gt [†]	Y: table_objects	Y	Y	arm,dual,ptr	arm,dual
KINEMATICS / BODY (8 implemented)								
edge.{foot_of, limb_adjacent, over_cell} (3)	edge	bias	given	N	Y	N	leg,hum	–
edge.{kin_child, mirror} (2)	edge	bias	given	N	Y	N	leg,hum,psi0	–
edge.kin_parent	edge	bias	given	N	Y	N	arm,dual,leg,hum,psi0	–
kin.ancestor	ancestor	bias	given	N	Y	N	arm,dual,leg,hum,psi0	–
kin.sibling	hop	bias	given	N	Y	N	arm,dual,leg,hum,psi0	–
PHYSICAL INTERACTION (5 implemented)								
ix.contact	bilinear	aug	probe, gt [†]	Y: grasp_target	Y	Y	arm,dual	arm,dual
ix.force_flow	flow	bias	probe, gt [†]	Y: stack	Y	N	arm,dual	arm,dual
ix.handover	bilinear	aug	probe, gt [†]	Y: grasp_target	Y	Y	dual	arm,dual
ix.held_by	bilinear	aug	probe, gt [†]	Y: grasp_target	Y	Y	arm,dual	arm,dual
ix.support	bilinear	aug	probe, gt [†]	Y: stack	Y	Y	arm,dual	arm,dual
TASK / PROCEDURE (15 implemented)								
edge.{actor_of, consumed_by, destination_of, enables, maintained, node_actor_of, output_to, patient_of, pred_arg, produced, role_in_event, role_points_to, support_of, target_of} (14)	edge	bias	given	N	Y	N	arm,dual	–
task.next_contact	bilinear	aug	probe, gt [†]	Y: reveal, surprise	Y	Y	arm,dual	arm,dual, mj-leg,wp-leg, simple
LOCOMOTION (2 implemented)								
leg.com_support	inert	readout	given	Y: terrain_steps	N	Y	leg,hum	mj-leg,wp-leg
leg.foothold	bilinear	aug	probe, gt [†]	Y: terrain_steps	Y	Y	leg,hum	mj-leg,wp-leg
UI (5 implemented)								
ui.above	order	bias	given	Y: cw_viewport, cw_depth	Y	N	ptr	cw
ui.drag_to	bilinear	aug	probe, gt [†]	Y: cw_viewport, cw_depth	Y	Y	ptr	none [‡]
ui.focus_next	edge	bias	given, gt [†]	Y: cw_viewport, cw_depth	Y	N	ptr	cw
ui.label_for	edge	bias	given, gt [†]	Y: cw_viewport, cw_depth, reveal, surprise	Y	N	ptr	cw
ui.same_window	same	bias	given	Y: cw_viewport, cw_depth	Y	N	ptr	cw
PROBE READOUTS (26 implemented)								
probe.arm.{acting_on, focused_on, goal_effect, held_by, looking_at, observed_effect, rel_pos, subtask, visible} (9)	inert	readout	given	N	N	Y	arm,dual	–
probe.legged.{contact, disp, fall, goal, subtask} (5)	inert	readout	given	N	N	Y	leg,hum	–
probe.pointer.{phase, rel, slot} (3)	inert	readout	given	N	N	Y	ptr	–
probe.psi0.{active_hand, base_cmd, base_disp, contact, grasp_face, grasp_pt, hand_dist, lift, target_pos} (9)	inert	readout	given	N	N	Y	psi0	–

Generated by make_figures.py from the registries at c18101e8.

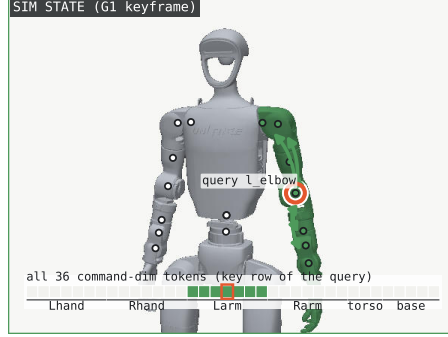
Table 6: Every implemented relation factor. 75 factors: 18 generate or label data, 46 act on attention logits or masks, 38 supervise a readout (several per factor possible). Rows with identical classification are merged (**edge.***, **probe.<net>.***; count in parentheses). gt[†] = gt (training only): simulator truth, never a deployable input. Data gen: a data generator is declared or a relgen label exists (scene parts upright, transforms italic); probe targets are task-pipeline packet labels, hence N. Attn: form ∈ {bias, aug, gate, mask}. Read: a readout is declared. Nets: families whose **resolve(family=...)** accepts the factor. Data envs: label caps and scene part available; [‡] no registered environment provides the caps yet. Fig. 5 draws one tile per factor.

Figure 5: Every implemented relation factor, one tile each (75 tiles on 10 pages, three per row, families in the row order of Table 6). Each tile shows one factor’s term for one query token (ring or outlined chip) on a real simulator or environment state, computed by the repository’s factor operators or label code; *nothing is trained attention or a trained probe output*. Frame colour = family (Fig. 2); probe readouts are coloured by what they encode. Top-left chip: scene source (SCRIPTED TEACHER states are privileged). Under each tile: the factor and its term source, ● *public*: computed from public inputs, deployable as shown; ▲ *gt train-only*: the colour is the simulator-truth label that trains a probe or readout, never a deployed input; ◆ *est-probe*: the deployed term is the probe’s estimate (not shown); then what the term encodes on the shown state, its operator and form (learned coefficients are *set by hand*, values stated) and source, and its equation. ILLUSTRATIVE: not fully computable from shipped code or data (reason under the tile). `make_figures.py` tiles.

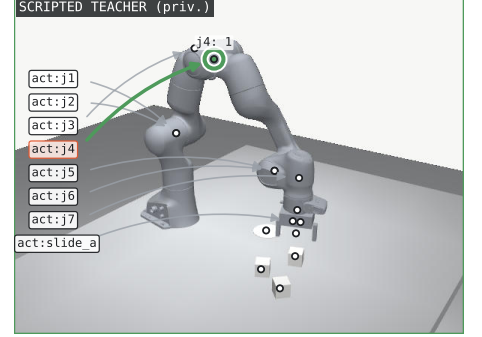
STRUCTURE / EDGES (9 implemented)



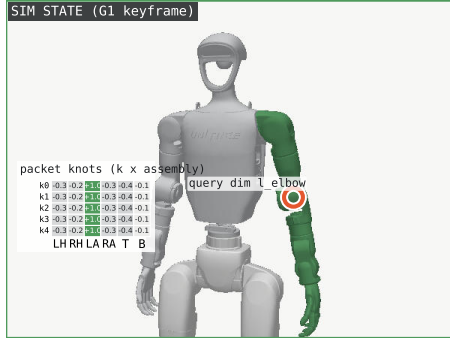
edge.node_in_assembly
a joint / sensor token attends to the assembly (arm, gripper) it belongs to. edge x bias at $act > ctx$ and $ctx > ctx$ (solid: act nodes, dashed: touch / grip sensors; $w=1$). Source: public morphology (featurizer).
 $\mathbb{1}[j = asm(i)]$



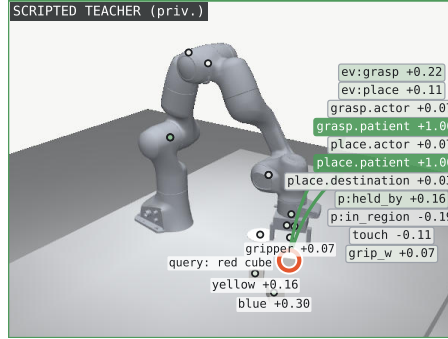
edge.same_assembly
a command dim attends to the other dims of its own assembly (here the left arm). edge x bias at $dims > dims$ (g1-dim-rel-v1 channel same_assembly; $w=1$). Source: public morphology (bodies.g1_simple).
 $\mathbb{1}[asm(i) = asm(j), i \neq j]$



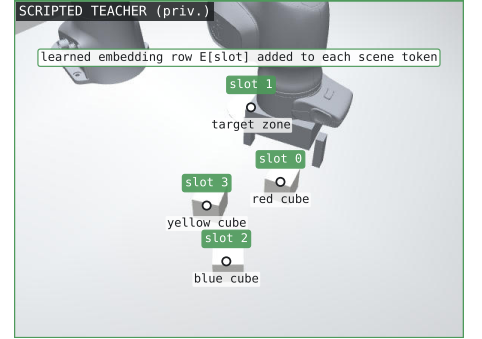
edge.same_node
an action-node query attends to its OWN morphology token (one per actuated joint). edge x bias, $w * A[i,j]$ at $act > ctx$ (arm-rel-v1, $w=1$ shown). Source: public morphology (featurizer).
 $\mathbb{1}[j = morph(i)]$



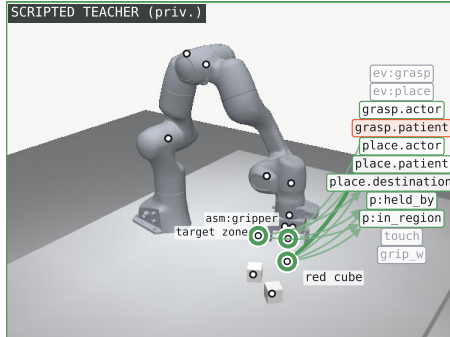
id.same_assembly
soft same-assembly term: a command dim prefers knots of its own assembly (left arm); other assemblies stay readable. same x aug, random 16-d codes for 64 ids (small cross-talk), $g=1$, $dims > knots$. Source: public morphology (bodies.g1_simple).
 $\langle e(a_i), e(a_j) \rangle \approx \mathbb{1}[a_i = a_j]$



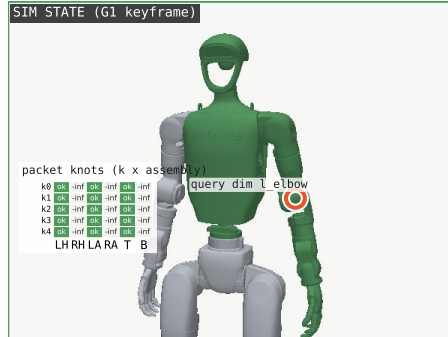
id.same_body
tokens denoting the SAME entity (a tracker slot and the task roles bound to it) attend to each other. same x aug: fixed random 16-d unit codes for 64 ids (1 on equal ids, small cross-talk otherwise), $g=1$. Source: public ids (featurizer pointers).
 $\langle e(id_i), e(id_j) \rangle \approx \mathbb{1}[id_i = id_j]$



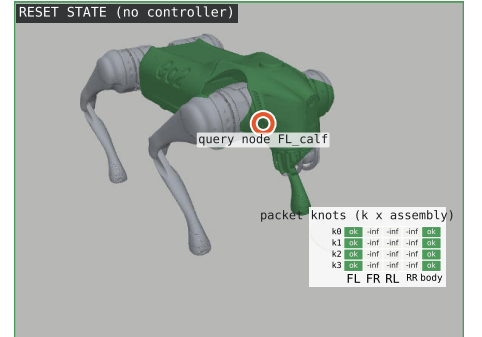
id.slot_handle
each scene token carries its public tracker slot address as a learned embedding, so a slot keeps its identity across steps. embed form (unary, no attention term); E learned. Source: public tracker slot ids.
 $x_k \leftarrow x_k + E[slot_k]$



msg.incidence
role and predicate tokens receive a message from the entity they point to (arrows: entity $\rightarrow$ role). message form, not a logit term; W learned; control 'serialized' = same facts as hashed text. Source: public task graph + tracker slots.
 $h_i \leftarrow h_i + W h_{ptr(i)}$

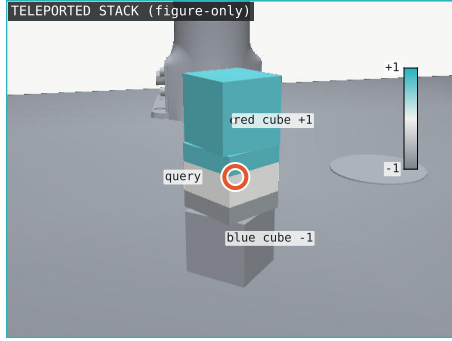


route.assembly_reads
system 0 (Psi0 Realizer): a command dim reads packet knots of its own assembly and its kinematic neighbours only (left arm: arm, hand, torso). same x mask with `params.reads = bodies.g1_simple.READS`. Source: public morphology (bodies.g1_simple).
0 if `READS[asm(i), asm(j)]` else $-\infty$



route.own_assembly
system 0 (legged Realizer): a joint node reads only the packet knots of its own limb plus the body assembly. same x mask on `assembly_id`, `params.also_key_field = body` (preset legged-s0). Source: public morphology (LeggedMorph).
0 if $a_j \in \{a_i, body\}$ else $-\infty$

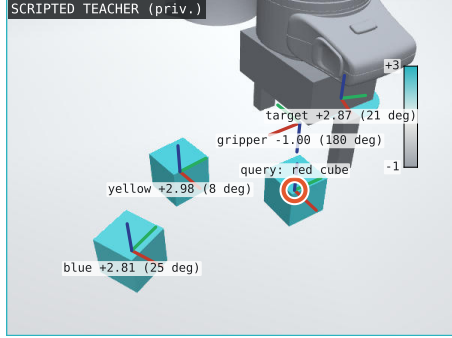
GEOMETRY (5 implemented)



geo.above

above / below along gravity: +1 for tokens (and surface points) higher than the query by > 1 cm, -1 lower, 0 in the dead zone (query: middle cube of a stack). order x bias on pos3d, axis +z, margin 0.01 m ($w=1$). Source: tracker pos3d (given).

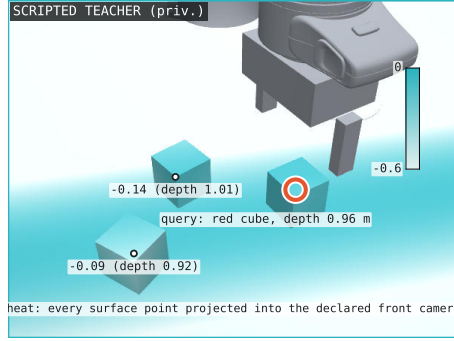
$$\text{sign}((r_j - r_i) \cdot \hat{z}) \text{ if } |\cdot| > 1 \text{ cm}$$



geo.orient

relative rotation between token frames: $\text{tr}(R_i^T R_j) = 1 + 2 \cos(\text{angle})$ (axes: x red, y green, z blue). rel_rot x aug, $B = I$ (learned per head). Source: gt orient label (sim truth, training-only); deployed = probe estimate (object frames are not tracked).

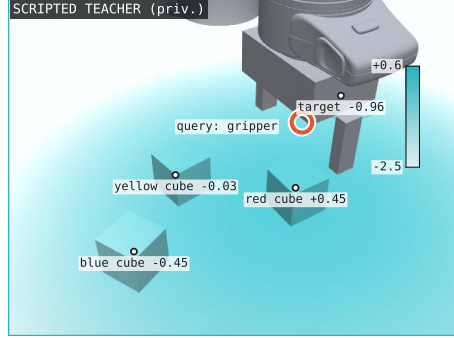
$$\langle R_i B, R_j \rangle_F = \text{tr}(R_i^T R_j), \quad B = I$$



geo.depth3d

PaPE over (u, v, depth) in the declared front camera; weight on depth only, so tokens at the query's depth score high. $\text{sqdiff} + \text{diff} \times \text{aug}$ on cam_uvd , $a=(0,0,60)$, $b=0$ (learned). Source: projected tracker pos3d (given); default source = probe.

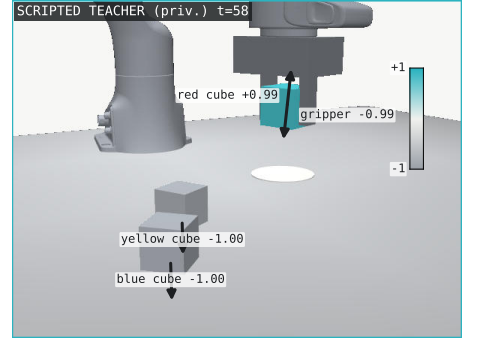
$$-(r_j - r_i)^T M(r_j - r_i), \quad r = (u, v, \text{depth})$$



geo.pos3d

PaPE relative-position term: tokens near and below the gripper score high; heat = the term at every surface point. $\text{sqdiff} + \text{diff} \times \text{aug}$, $a=30$, $b=(0,0,-8)$, $W_p=I$ (learned per head). Source: public FK + tracker pos3d (given).

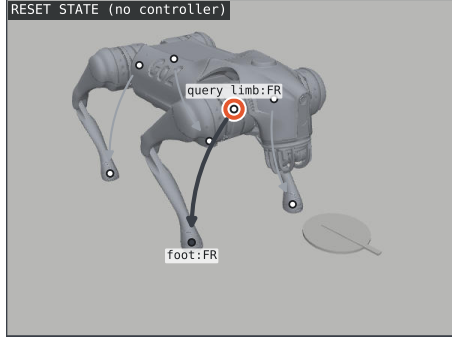
$$-(r_j - r_i)^T M(r_j - r_i) + \tilde{b}^T (r_j - r_i)$$



geo.normal_align

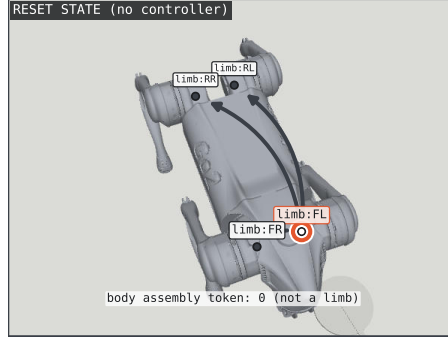
query direction b vs each token's contact normal (arrows: outward normal, mean over the token's contacts). align x aug, world frame, $b=(0,0,1)$ (learned per head). Source: gt contact_normal (sim truth, training-only); deployed = probe. $\langle b_i, n_j \rangle, \quad b = (0,0,1)$

Figure 5: Every implemented relation factor (continued, page 2/10). Chips, glyphs and colours as on the first page.

**edge.foot_of**

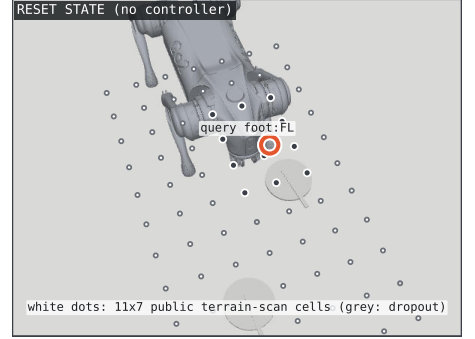
a limb token attends to its own foot token (feet exist only on foot-carrying assemblies; limbs drawn at hips, feet at foot sites). edge x bias at $\text{ctx} > \text{ctx}$ (legged-rel-v1; $w=1$). Source: public morphology (LeggedMorph).

$$\mathbb{1}[j = \text{foot}(i)]$$

**edge.limb_adjacent**

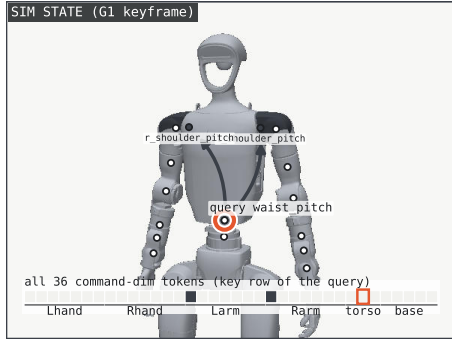
a limb token attends to the other limbs of the same kind on the trunk (all four legs; tokens drawn at their hips). edge x bias at $\text{ctx} > \text{ctx}$ (legged-rel-v1, symmetric; $w=1$). Source: public morphology (LeggedMorph).

$$\mathbb{1}[\text{kind}(i) = \text{kind}(j) \neq \text{body}, i \neq j]$$

**edge.over_cell**

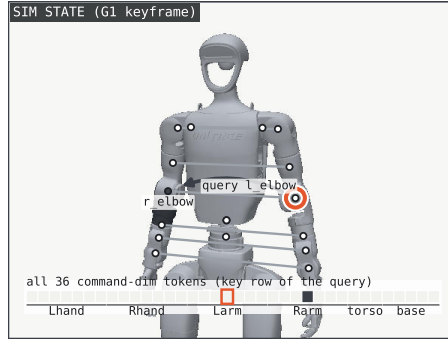
a foot token attends to the terrain-scan cells of its landing window (0.05 m behind to 0.35 m ahead, ± 0.15 m aside of its default stance, body frame). edge x bias at $\text{ctx} > \text{ctx}$ (legged-rel-v1; $w=1$). Source: public morphology + public terrain scan.

$$\mathbb{1}[\text{cell } j \in \text{window}(\text{foot } i)]$$

**edge.kin_child**

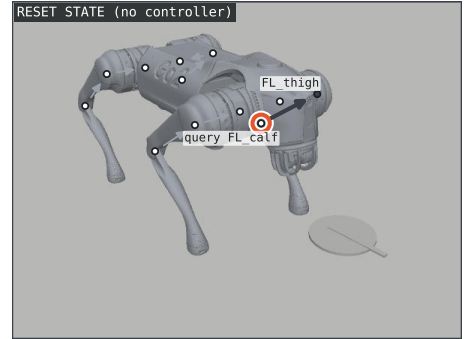
a command dim attends to its kinematic children (waist pitch $\rightarrow$ both shoulder pitches). edge x bias at $\text{dims} > \text{dims}$ (channel kin_child; $w=1$). Source: public morphology (bodies.g1_simple).

$$\mathbb{1}[\text{parent}(j) = i]$$

**edge.mirror**

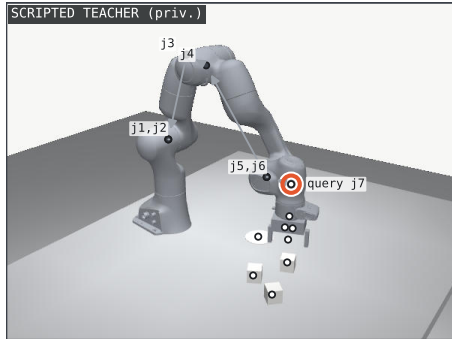
a command dim attends to its left/right homologue (grey lines: every mirror pair of the body). edge x bias at $\text{dims} > \text{dims}$ (channel mirror; $w=1$). Source: public morphology (bodies.g1_simple).

$$\mathbb{1}[j = \text{mirror}(i)]$$

**edge.kin_parent**

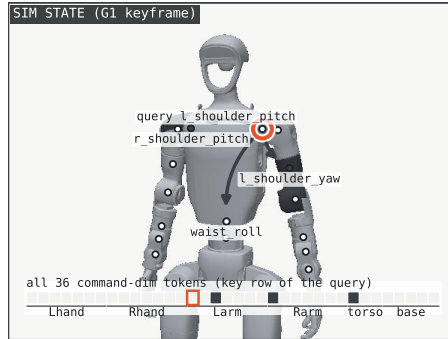
a joint token attends to its kinematic parent joint (grey: every parent edge of the four legs). edge x bias at $\text{ctx} > \text{ctx}$ (legged-rel-v1, from the per-assembly depth chain; $w=1$). Source: public morphology (LeggedMorph).

$$\mathbb{1}[j = \text{parent}(i)]$$

**kin.ancestor**

a joint token attends to every joint up its kinematic chain (transitive closure of kin_parent; grey arrows: the parent edges). ancestor (closure) x bias at $\text{ctx} > \text{ctx}$ (arm-rel-v1; $w=1$). Source: public morphology (featurizer).

$$\mathbb{1}[j \text{ ancestor of } i]$$

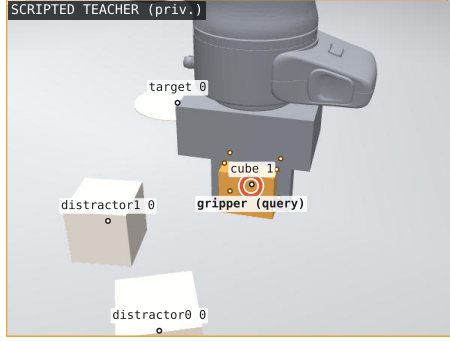
**kin.sibling**

dims at undirected tree distance 2: the sibling ($r_shoulder_pitch$) and, by the same op, grandparent and grandchild. hop x bias over the symmetrized kin_parent graph (hops=2; $w=1$). Source: public morphology (bodies.g1_simple).

$$\mathbb{1}[d_{\text{tree}}(i, j) = 2]$$

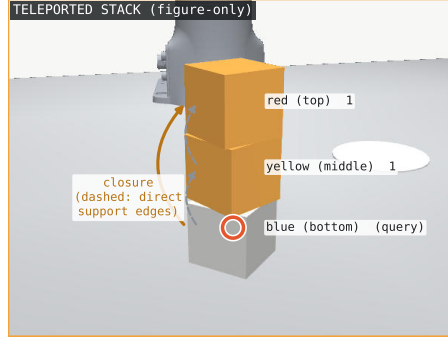
Figure 5: Every implemented relation factor (continued, page 3/10). Chips, glyphs and colours as on the first page.

PHYSICAL INTERACTION (5 implemented)



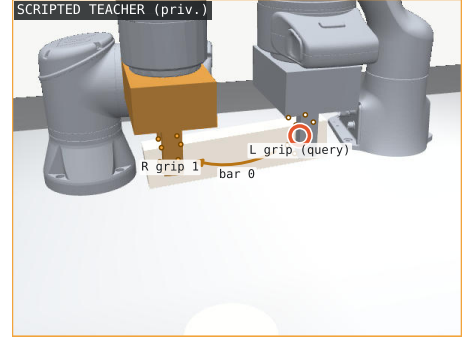
ix.contact

two entities in geometric contact (sim contact truth): query gripper $\rightarrow$ cube 1, distractors / target 0. White dots: the gripper-cube contact points. bilinear $\times$ aug (probe). Source: shown: gt label (training-only); deployed: estimated-probe.
contact(i, j)



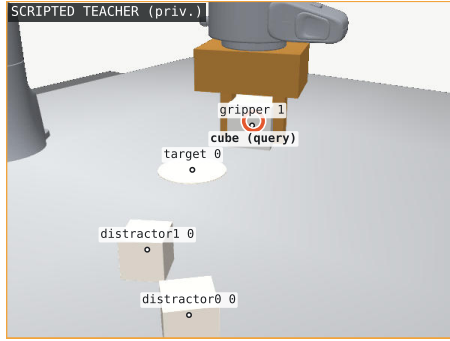
ix.force_flow

transitive closure of the support graph: all the query carries, directly or via others. Bottom cube $\rightarrow$ middle 1, top 1 (direct edge only to the middle). flow (closure) $\times$ bias over support-v1. Source: shown: closure of gt support (training-only); deployed: closure of the ix.support est-probe.
flow⁺(support)(i, j)



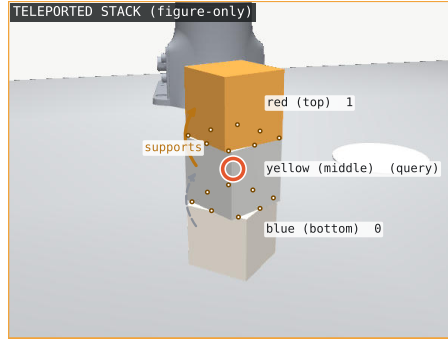
ix.handover

two manipulator assemblies in contact with the same object at once (a handover in progress): query L grip $\rightarrow$ R grip 1, bar / target 0. Dots: bar contacts of each hand. Dual env only. bilinear $\times$ aug (probe). Source: shown: gt label (training-only); deployed: estimated-probe.
handover(i, j)



ix.held_by

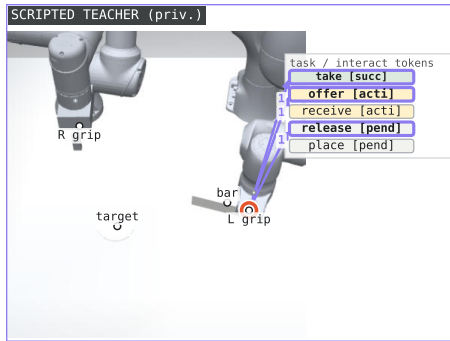
object held by a manipulator assembly (≥ 2 contacts with its hand bodies): lifted cube $\rightarrow$ gripper 1; target and distractors 0. bilinear $\times$ aug (probe). Source: shown: gt label (training-only); deployed: estimated-probe.
held_by(i, j)



ix.support

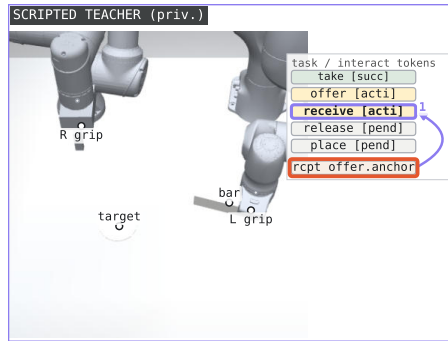
a supports b: contact + normal within 30° of gravity-up at a's top. Query middle cube $\rightarrow$ top 1 (row); the bottom cube supports it (grey arrow, column). Emits the estimated support-v1 graph. bilinear $\times$ aug (probe, emits). Source: shown: gt label (training-only); deployed: est-probe.
support(i, j)

TASK / PROCEDURE (15 implemented)



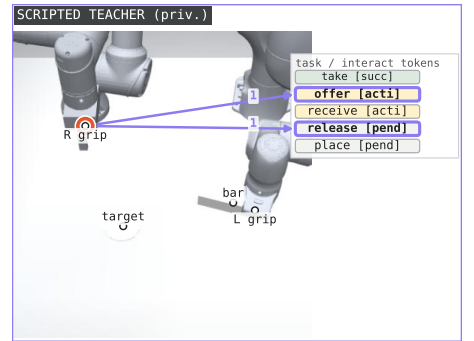
edge.actor_of

manipulator entity $\leftrightarrow$ every event it is (cooperating) actor of; L grip acts in take, offer, release. Typed edge channel of the arm-rel-v1 graph, one learned weight per head. edge $\times$ bias. Source: public (given): supplied task graph + runtime.
 $\mathbb{1}[\{i, j\} \in E_{actor_of}]$



edge.consumed_by

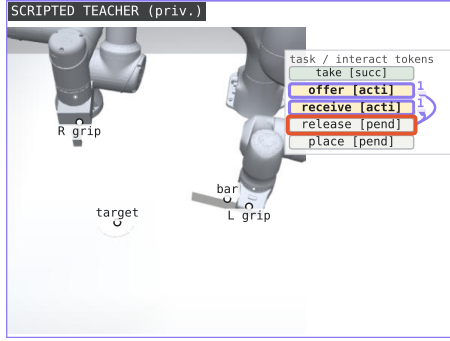
runtime receipt $\leftrightarrow$ the event whose output-binding role consumes it: offer.anchor is consumed by receive. edge $\times$ bias. Source: public (given): supplied task graph + runtime.
 $\mathbb{1}[\{i, j\} \in E_{consumed_by}]$



edge.destination_of

destination entity $\leftrightarrow$ event: the right gripper is where 'offer' and 'release' deliver the bar (target zone is place's destination). edge $\times$ bias. Source: public (given): supplied task graph + runtime.
 $\mathbb{1}[\{i, j\} \in E_{destination_of}]$

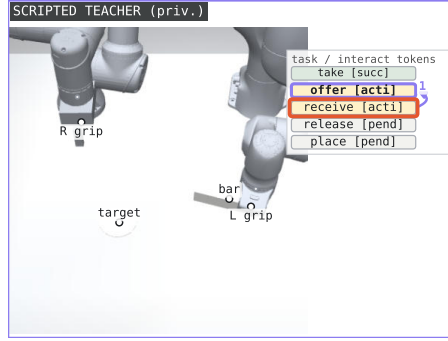
Figure 5: Every implemented relation factor (continued, page 4/10). Chips, glyphs and colours as on the first page.



edge.enables

event $\rightarrow$ event it requires COMPLETED (requires_completed): release needs receive and offer done. Public task-graph precedence. edge $\times$ bias. Source: public (given): supplied task graph + runtime.

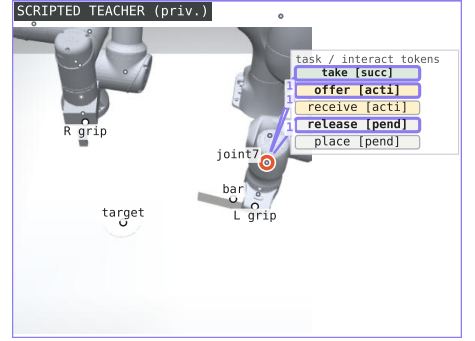
$$\mathbb{1}[j \in \text{requires_completed}(i)]$$



edge.maintained

event $\rightarrow$ event it requires ACTIVE (requires_active): receive is only valid while offer is held. edge $\times$ bias. Source: public (given): supplied task graph + runtime.

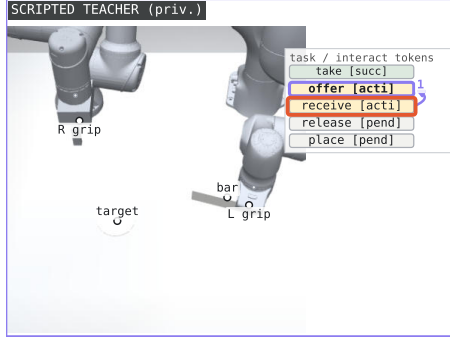
$$\mathbb{1}[j \in \text{requires_active}(i)]$$



edge.node_actor_of

action node (one joint-command token) $\rightarrow$ every event its manipulator acts in: a 2-hop composition node $\rightarrow$ assembly $\rightarrow$ actor binding. Left-arm joint7 $\rightarrow$ take, offer, release. edge $\times$ bias (act>ctx). Source: public (given): supplied task graph + runtime.

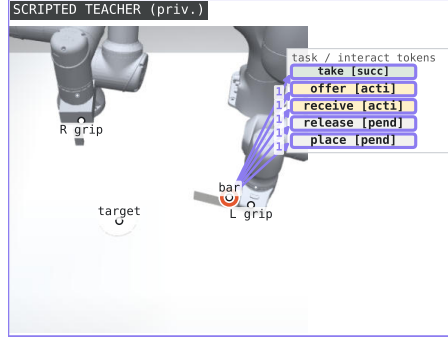
$$\mathbb{1}[\text{asm}(i) \in \text{actors}(j)]$$



edge.output_to

event $\leftrightarrow$ event consuming its output: receive's reference role binds offer's 'anchor' output (event_output binding). edge $\times$ bias. Source: public (given): supplied task graph + runtime.

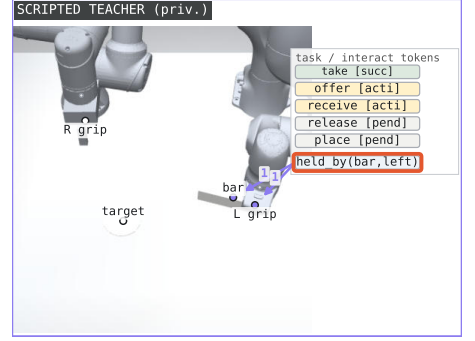
$$\mathbb{1}[\{i, j\} \in E_{\text{output_to}}]$$



edge.patient_of

patient entity $\leftrightarrow$ the events that act on it: the bar is the patient of all five handover events (take, offer, receive, release, place). edge $\times$ bias. Source: public (given): supplied task graph + runtime.

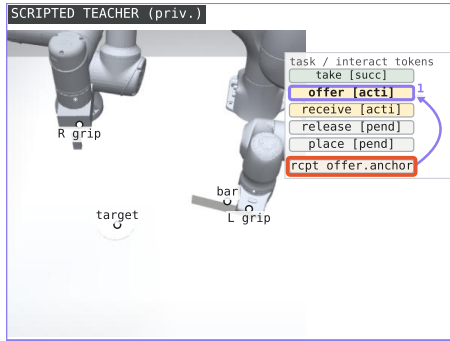
$$\mathbb{1}[\{i, j\} \in E_{\text{patient_of}}]$$



edge.pred_arg

predicate-estimate token $\leftrightarrow$ its argument entities: held_by(bar, left) points at the bar slot and the left-gripper assembly. edge $\times$ bias. Source: public (given): supplied task graph + runtime.

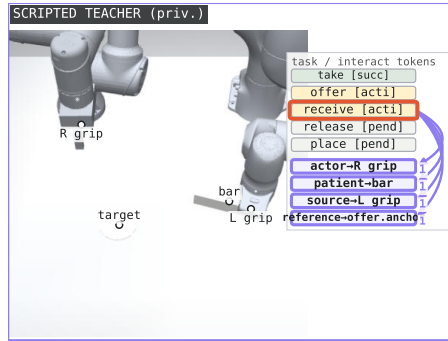
$$\mathbb{1}[\{i, j\} \in E_{\text{pred_arg}}]$$



edge.produced

runtime receipt $\leftrightarrow$ the event that produced it: the 'anchor' receipt (interact bank) was emitted by offer. edge $\times$ bias. Source: public (given): supplied task graph + runtime.

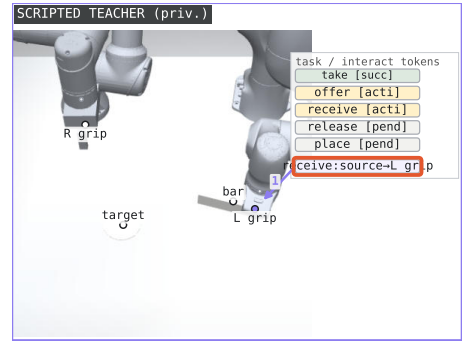
$$\mathbb{1}[\{i, j\} \in E_{\text{produced}}]$$



edge.role_in_event

role-slot token $\leftrightarrow$ its event: receive owns four ordered role slots (actor, patient, source, reference); multiplicity is explicit. edge $\times$ bias. Source: public (given): supplied task graph + runtime.

$$\mathbb{1}[\{i, j\} \in E_{\text{role_in_event}}]$$



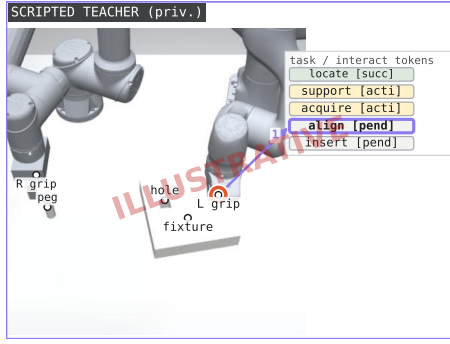
edge.role_points_to

role-slot token $\rightarrow$ the entity token it is bound to: receive:source $\rightarrow$ left-gripper assembly (an output binding points to no entity). edge $\times$ bias. Source: public (given): supplied task graph + runtime.

$$\mathbb{1}[j = \text{bound}(i)]$$

Figure 5: Every implemented relation factor (continued, page 5/10). Chips, glyphs and colours as on the first page.

TASK / PROCEDURE (15 implemented, continued)

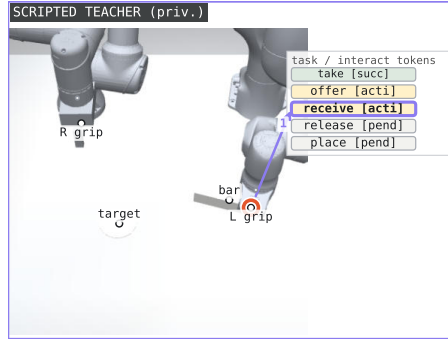


edge.support_of

support-role entity $\leftrightarrow$ event: the left arm braces the fixture during align. edge $\times$ bias. Source: illustrative: public edge on a figure-only task-graph variant.

$$\mathbb{1}[\{i, j\} \in E_{\text{support_of}}]$$

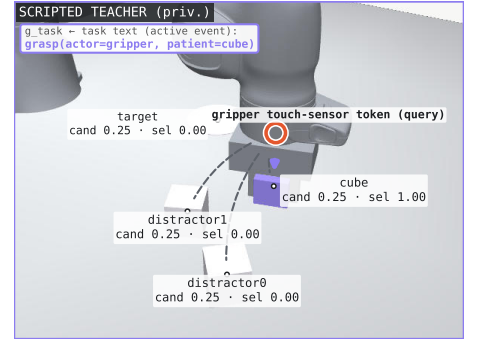
ILLUSTRATIVE: no shipped arm or dual task graph binds a support role to an entity; align:support is rebound to 'left' for this figure only.



edge.target_of

target entity (and every role without its own channel, here receive:source) $\leftrightarrow$ event: the left gripper is the source of 'receive'. edge $\times$ bias. Source: public (given): supplied task graph + runtime.

$$\mathbb{1}[\{i, j\} \in E_{\text{target_of}}]$$

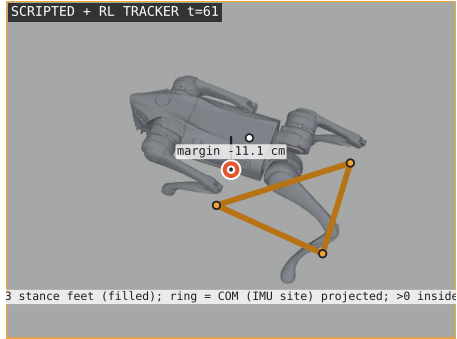


task.next_contact

manipulator $\rightarrow$ graspable score, sharpened by the task gate ($g_{\text{task}} = 1.00$ at init, learned). Dashed: public candidate edges ($1/|E|$); solid: selected (gt reveal posterior given contact). bilinear $\times$ aug, gated. Deployed: est-probe $\times$ g_{task} .

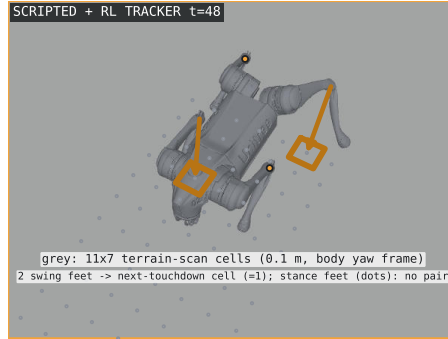
$$g_{\text{task}} \text{ next}(i, j)$$

LOCOMOTION (2 implemented)



leg.com_support

readout of the stability margin: signed planar distance of the COM projection to the stance-foot support polygon (m, + inside). readout (inert), address assembly, gauss (1-d). Source: gt sim truth (contacts), training-only. signed dist(COM_{xy} , hull(stance feet))

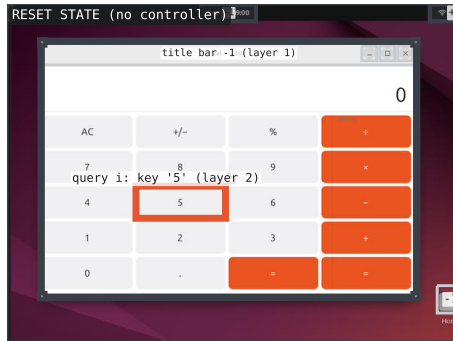


leg.foothold

pair label: each swinging foot $\rightarrow$ the terrain-scan cell containing its next touchdown (hindsight); planted feet have no true pair. op bilinear, form aug (the bias IS the pair probe). Source: gt label (future touchdown), training-only; deployed = estimated-probe. foothold(foot_f , cell_c)

Figure 5: Every implemented relation factor (continued, page 6/10). Chips, glyphs and colours as on the first page.

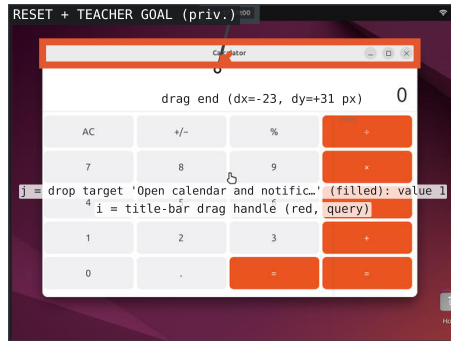
UI (5 implemented)



ui.above

signed render order: +1 if widget j is drawn above i (higher dense z-layer), -1 below, 0 same layer; dark = +1 (resize handles, shell), light = -1 (window frame, desktop icon). op order over field zlayer, form bias (antisymmetric). Source: public (given widget tree).

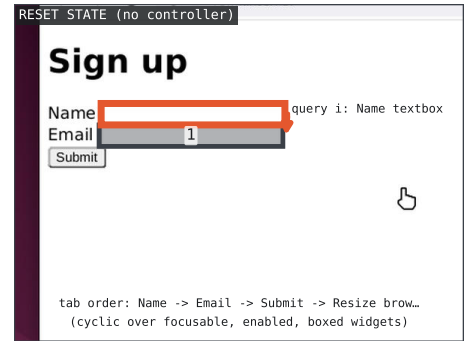
$\text{sign}(z_j - z_i)$



ui.drag_to

pair label: the teacher's drag handle -> the nearest widget outside the dragged window to where the drag ends (cw/drag_window); 1 pair of 2116 valid. op bilinear, form aug (the bias IS the pair probe). Source: gt label (teacher goal), training-only; deployed = estimated-probe.

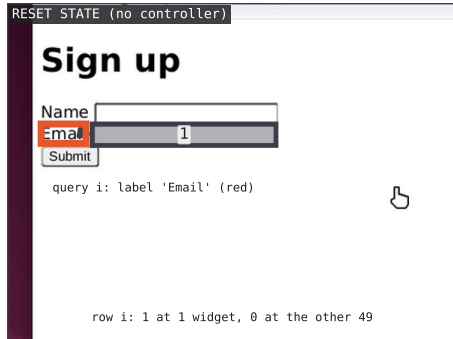
$\text{drag_to}(i, j)$



ui.focus_next

public UI edge: i, j consecutive in the tab order (focusable, enabled widgets in scene order, cyclic); Name -> Email. op edge, form bias (edges:ui-rel-v1 channel focus_next). Source: public (given widget tree).

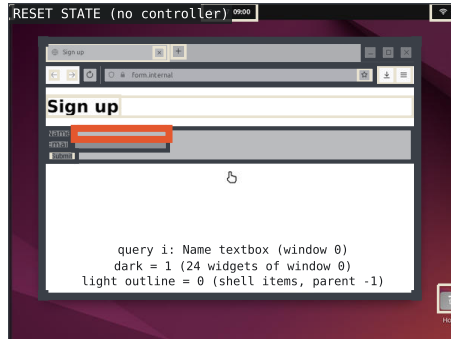
$\text{focus_next}(i, j)$



ui.label_for

public UI edge: a role=label widget -> the next widget after it in the same window (scene order); here 'Email' -> its textbox. op edge, form bias (edges:ui-rel-v1 channel label_for). Source: public (given widget tree).

$\text{label_for}(i, j)$



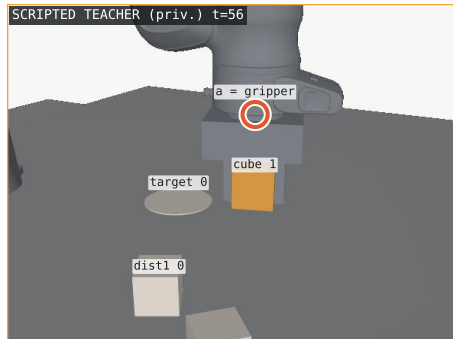
ui.same_window

1 when two widgets belong to the same window (parent_id equality; shell items such as the tab strip have parent -1 and never match). op same over field parent_id, form bias. Source: public (given widget tree).

$\mathbb{1}[\text{parent}_i = \text{parent}_j]$

PROBE READOUTS (26 implemented)

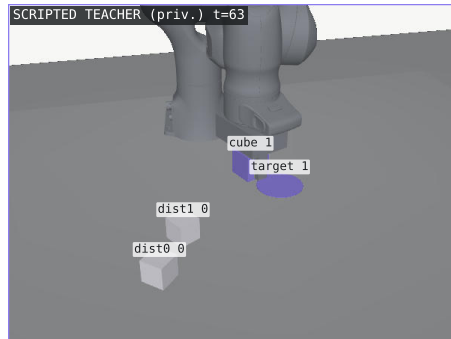
colour = what the readout encodes: cyan geometric, amber physical interaction, violet task / procedure



probe.arm.acting_on

readout of whether assembly a's hand links touch entity e (contact truth). readout (inert), address entity x assembly, bce. Source: gt sim truth (Session.truth held_by / contacts), training-only.

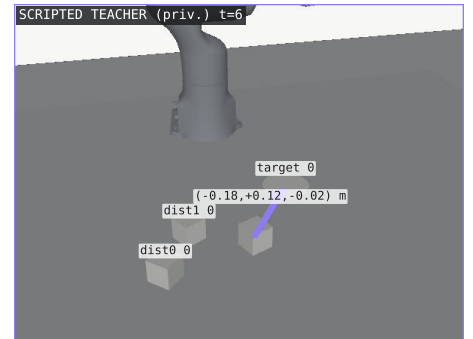
$\text{contact}(e, a)$



probe.arm.focused_on

readout of which entities a currently ACTIVE task event binds (active op 'place': cube + target). readout (inert), address entity, bce. Source: public label (task runtime status + bindings).

$\text{focus}(e) = \exists \text{ active event } \ni e$



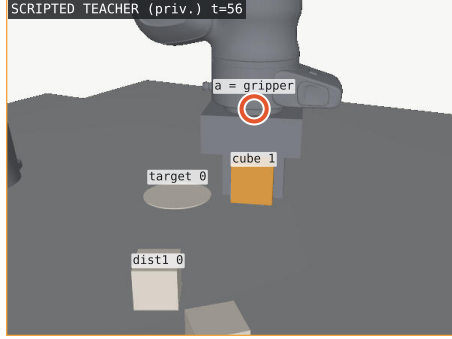
probe.arm.goal_effect

readout of the displacement the task asks of each entity: destination minus patient for open events (cube -> target; others 0). readout (inert), address entity, gauss. Source: public label (task spec + public pose estimates).

$\hat{p}_{\text{dst}} - \hat{p}_{\text{patient}}$ (open events)

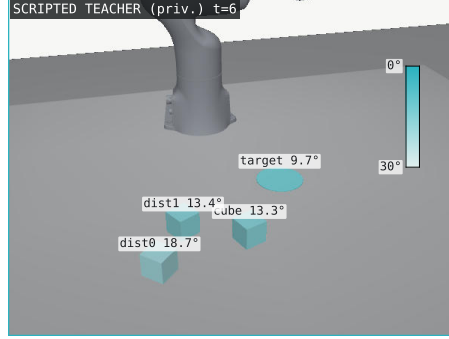
Figure 5: Every implemented relation factor (continued, page 7/10). Chips, glyphs and colours as on the first page.

PROBE READOUTS (26 implemented, continued)



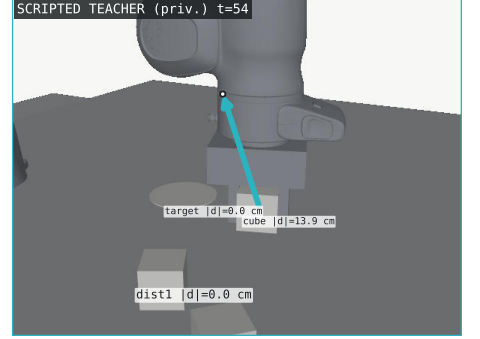
probe.arm.held_by

readout of whether entity e is held by assembly a (here gripper; cube 1, others 0). readout (inert), address entity x assembly, bce. Source: gt sim truth (Session.truth held_by / contacts), training-only.
 $\text{held}(e, a)$



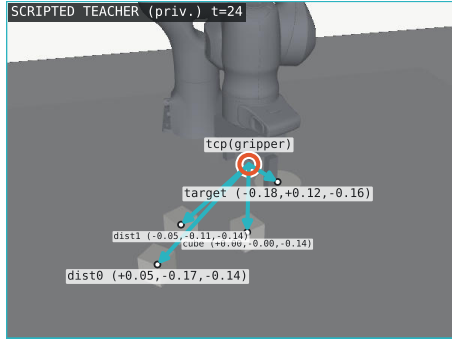
probe.arm.looking_at

readout of each entity's angle off the front-camera optical axis (gaze label, degrees, scaled 1/30). readout (inert), address entity, mse. Source: gt sim truth (camera pose + body pose), training-only.
 $\angle(\text{cam axis}, p_e - c) [\text{deg}]$



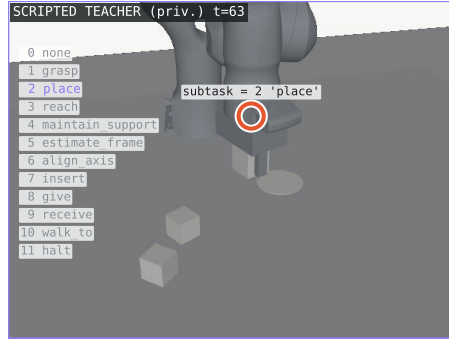
probe.arm.observed_effect

readout of each entity's realized displacement over the 16-tick chunk (future_disp; cube lifted, others 0). readout (inert), address entity, gauss. Source: gt sim truth (future state), training-only.
 $p_e(t+H-1) - p_e(t)$, $H=16$



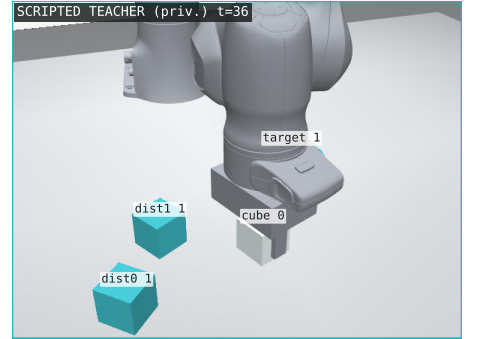
probe.arm.rel_pos

readout of entity position relative to the assembly's TCP, (x,y,z) m, world frame. readout (inert), address entity x assembly, gauss (μ , logvar). Source: gt sim truth, training-only.
 $p_e - p_{\text{tcp}}(a) [\text{m}]$



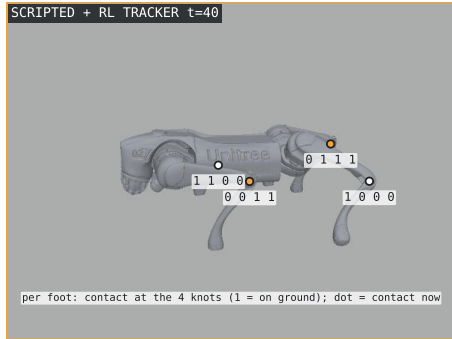
probe.arm.subtask

readout of the active task operator for the assembly (12-way; this episode: grasp -> place -> none). readout (inert), address assembly, ce. Source: public label (task runtime: first ACTIVE event's operator).
 $\text{arg op}(\text{active event})$



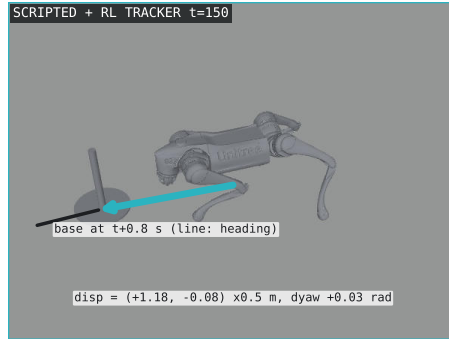
probe.arm.visible

readout of whether each entity is visible in the front camera (cube occluded by the closing gripper here: 0). readout (inert) on the packet, address entity, bce. Source: gt sim truth, training-only target; deployed = probe estimate.
 $\text{visible}(e) \in \{0, 1\}$



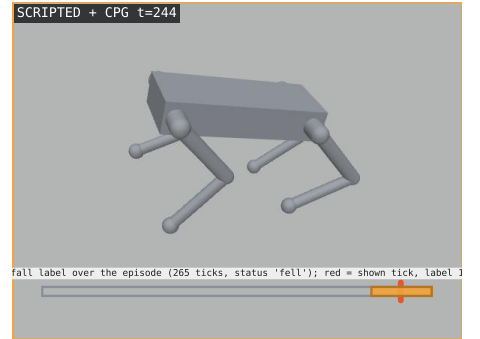
probe.legged.contact

readout of each foot's ground contact at the 4 packet knots (+0.1/0.3/0.5/0.7 s) of a trot. readout (inert), address knot x assembly (feet), bce. Source: gt sim truth (future contacts), training-only.
 $c_f(t + \tau_k)$, $\tau = 0.1, 0.3, 0.5, 0.7 \text{ s}$



probe.legged.disp

readout of the base displacement over the next 0.8 s in the start body frame (xy in 0.5 m units, yaw change in rad). readout (inert), address assembly, gauss (3-d). Source: gt sim truth (future base pose), training-only.
 $R_{\psi_t}^\top (p_{t+H} - p_t) / 0.5$, $\Delta\psi$

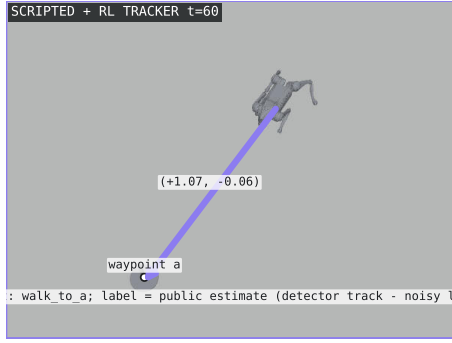


probe.legged.fall

readout of an imminent fall: 1 in the last 0.8 s of an episode that ends in a fall (non-foot ground contact, low base or tilt). readout (inert), address assembly, bce. Source: gt sim truth (episode outcome), training-only.
 $\mathbf{1}[\text{fell}] \cdot \mathbf{1}[T_{\text{end}} - t \leq 0.8 \text{ s}]$

Figure 5: Every implemented relation factor (continued, page 8/10). Chips, glyphs and colours as on the first page.

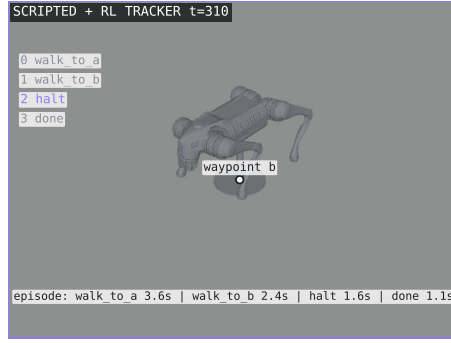
PROBE READOUTS (26 implemented, continued)



probe.legged.goal

readout of the active event's goal entity (here the waypoint) in the body yaw frame, /2 m, as the PUBLIC context estimates it. readout (inert), address assembly (body row), gauss (2-d). Source: public label (public context estimate).

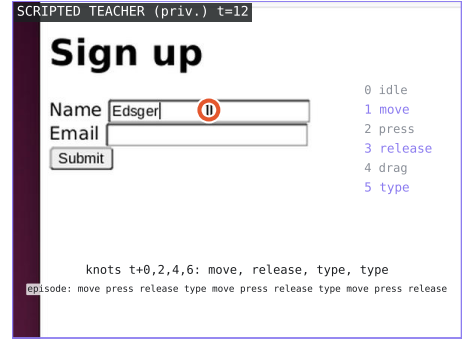
$goal_{active}$ in body yaw frame / 2 m



probe.legged.subtask

readout of which task event is active (walk_to_a, walk_to_b, halt, done) from the public task runtime. readout (inert), address assembly, ce. Source: public label (task runtime).

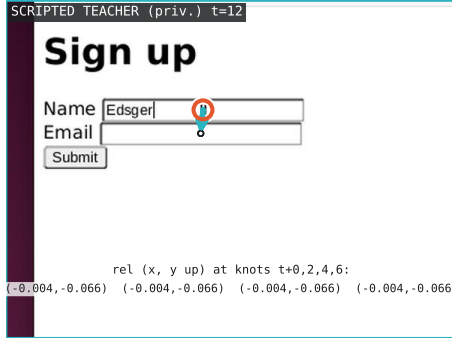
index of the active task event (4-way)



probe.pointer.phase

readout of the interaction phase of the commanded tick (idle, move, press, release, drag, type) at each knot. readout (inert), address knot x assembly, ce. Source: gt teacher command, training-only.

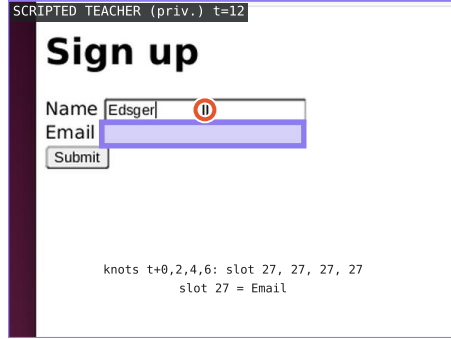
phase of the commanded tick (6-way)



probe.pointer.rel

readout of the target point relative to the pointer, normalized screen units (x right, y up), per knot. readout (inert), address knot x assembly, gauss (2-d mean + logvar). Source: gt teacher target, training-only.

$xy_{target}(t+j) - xy_{ptr}(t)$ [screen/2]



probe.pointer.slot

readout of WHICH widget the current plan step targets, as its stable slot index (opaque codes only, no widget content). readout (inert), address knot x assembly, ce over 80 slots. Source: gt teacher target, training-only.

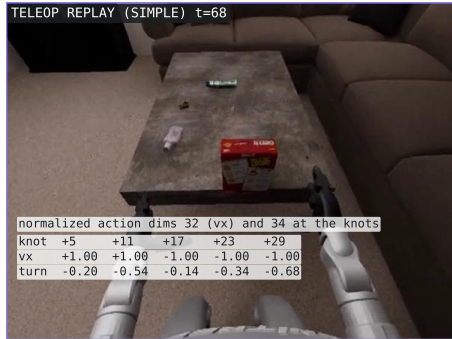
slot index of the step's target widget (80-way)



probe.psi0.active_hand

readout of which hand the packet binds to the target (here right); -1 (no contact ever) is masked. readout (inert), address body (per packet), ce over 2. Source: gt sim truth (future contacts), training-only.

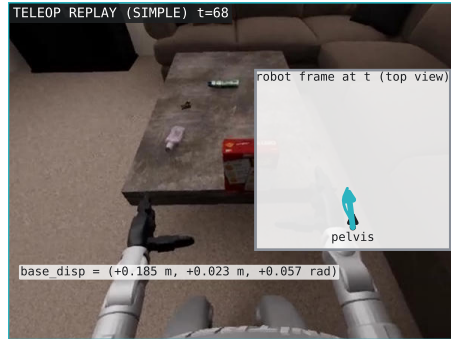
hand bound to the target over the packet



probe.psi0.base_cmd

readout of the demonstrated base command at each knot (normalized action dims 32 = vx, 34 = turning flag per bodies.g1_simple). readout (inert), address knot x pair (slot 0), gauss (2-d). Source: label from the recorded demonstration actions, training-only.

demonstrated (vx, turn) command at knot k



probe.psi0.base_disp

readout of the pelvis displacement over the packet ($t \rightarrow t+29$) in the robot frame at t , plus yaw change. readout (inert), address body, gauss (3-d). Source: gt sim truth (pelvis pose), training-only.

pelvis ($\Delta x, \Delta y, \Delta \psi$) over 30 ticks



probe.psi0.contact

readout of hand-target contact per hand at the 5 knots (here the right hand closes on the target). readout (inert), address knot x pair, bce. Source: gt sim truth (recorded contacts), training-only.

$\mathbb{1}[\text{hand}_h \text{ touches target}](t+k)$

Figure 5: Every implemented relation factor (continued, page 9/10). Chips, glyphs and colours as on the first page.

PROBE READOUTS (26 implemented, continued)



probe.psi0.grasp_face

readout of the object face (+x,-x,+y,-y,+z,-z) of each hand's first contact. readout (inert), address knot x pair (last knot), ce over 6. Source: gt sim truth, training-only (default OFF).

$2 \arg \max |p| + [p_{ax} < 0]$ (6 faces)

ILLUSTRATIVE: scene real; the label is not computable here (no contact positions in any recorded replay; needs SIMPLE / Isaac Sim).

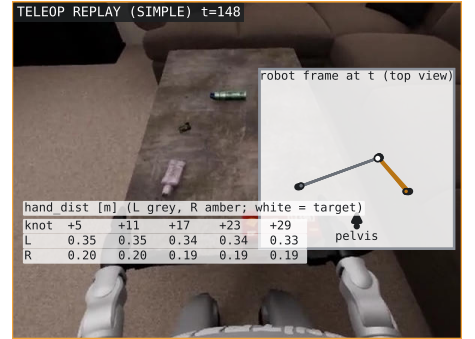


probe.psi0.grasp_pt

readout of each hand's FIRST contact point on the target in the object frame (last knot). readout (inert), address knot x pair (last knot), gauss (3-d). Source: gt sim truth, training-only (default OFF).

$R(q)^T (p_{\text{contact}} - p_{\text{obj}})$

ILLUSTRATIVE: scene real; the label is not computable here (no contact positions in any recorded replay; needs SIMPLE / Isaac Sim).



probe.psi0.hand_dist

readout of each hand's palm-to-target distance at the 5 packet knots (+5..+29 ticks of 20 ms). readout (inert), address knot x pair (pair = L/R hand), gauss. Source: gt sim truth (recorded SIMPLE replay), training-only.

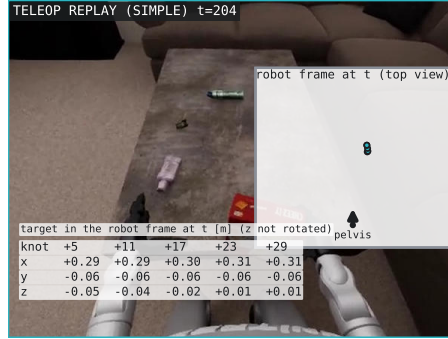
$\|p_{\text{palm},h}(t+k) - p_{\text{target}}(t+k)\|$



probe.psi0.lift

readout of whether the target is lifted ≥ 3 cm above its initial height at each knot. readout (inert), address knot x pair (slot 0), bce. Source: gt sim truth (target pose), training-only.

$\mathbb{1}[z_{\text{target}}(t+k) \geq z_0 + 3 \text{ cm}]$



probe.psi0.target_pos

readout of the target's position at each knot in the robot (pelvis yaw) frame at t. readout (inert), address knot x pair (slot 0), gauss (3-d). Source: gt sim truth (target pose), training-only.

$R_{\psi_t}^T (p_{\text{target}}(t+k) - p_{\text{pelvis}}(t))$

Figure 5: Every implemented relation factor (continued, page 10/10). Chips, glyphs and colours as on the first page.

Intended entry	Status	Catalog family	Operator $\times$ form	Needs (label caps; scene part)	Envs
STRUCTURE / EDGES (4 planned)					
<i>id.same_material</i>	planned (W2)	same-X symmetry / lattice; equivalence segmentation hierarchy	same $\times$ aug	materials ; needs material ids	all
<i>id.same_part_class</i>	planned (P)		same + edge	–	simple, arm
<i>id.same_segment</i>	planned (P)		same	needs image-patch tokens with masks	arm
<i>id.scene_parent</i>	planned (W2)		ancestor + hop	–	arm, legged, simple
GEOMETRY (3 planned)					
<i>geo.extent</i>	planned (P)	shape and scale	unary + sim	extent	arm, dual
<i>geo.occludes</i>	planned (P)	visibility	unary + edge	–	arm, dual
<i>geo.surface_normal</i>	planned (W2)	surface directions	align $\times$ aug	surface_normal	arm, dual
KINEMATICS / BODY (1 planned)					
<i>kin.cross_morphology</i>	planned (P)	mirror / morphology correspondence	same + edge	–	simple
PHYSICAL INTERACTION (6 planned)					
<i>ix.articulation</i>	planned (W2)	attachment / articulation	bilinear + align	articulation_pairs ; needs a drawer / door part	arm
<i>ix.collision_seq</i>	planned (W2)	collisions over time	flow	–	arm, dual, legged
<i>ix.contains</i>	planned (W2)	region topology; containment of content	bilinear + edge	contains_pairs ; needs bin part; bin part	arm, dual, cw
<i>ix.impulse_flow</i>	planned (W2)	force transfer	flow $\times$ bias	support_pairs , contact_forces	arm, dual
<i>ix.mating</i>	planned (W2)	mating / insertion	bilinear + align	mating_pairs ; support_insert task exists	arm
<i>ix.tool_target</i>	planned (W2)	tool use	bilinear $\times$ gate	tool_pairs ; needs a tool part	arm
TASK / PROCEDURE (3 planned)					
<i>rel.task_relevance</i>	planned (W2)	relevance / salience	unary $\times$ gate, readout	–	arm, dual, cw
<i>task.cause_effect</i>	planned (W2)	causality	edge + flow	labels from task receipts + interventions	arm, dual
<i>task.precedes</i>	planned (W2)	prerequisites	ancestor	–	arm, dual
TIME (3 planned)					
<i>time.co_motion</i>	planned (P)	motion	sim + diff	velocity labels exist	arm, dual, legged
<i>time.phase_sync</i>	planned (W2)	periodicity / phase	rel_rot + align	legged gait legs	legged
<i>time.same_track</i>	planned (registry)	correspondence (declared)	same $\times$ aug	–	–
PROPERTIES / AFFORDANCE / LANGUAGE (8 planned)					
<i>aff.<verb></i>	planned (P)	affordance scores	unary $\times$ readout	labels need per-object scripted affordance tables	arm, dual
<i>aff.affords_at</i>	planned (W2)	affordance pairs	bilinear $\times$ gate	–	arm, cw
<i>aff.compatible</i>	planned (P)	compatibility	bilinear + sim $\times$ gate	–	–
<i>aff.safe_region</i>	planned (P)	safety regions	unary	–	arm
<i>lang.refers_to</i>	planned (W2)	text grounding	bilinear	needs instruction tokens in the arm ctx	–
<i>lang.sim</i>	planned (P)	semantic similarity	sim	–	–
<i>prop.*</i>	planned (P)	physical properties	unary + sim	inertia, materials ; labels exist in MuJoCo:	arm, dual
<i>prop.vel</i>	planned (P)	fields and flows	unary	inertia, materials	–
OUT OF SCOPE (3 families; no physics or sensing for them in our environments): fields and flows; fluids, thermal, electrical, magnetic, optical, chemical; mechanics.					

Generated by `make_figures.py` from `research/relations_catalog.md` and the registries at `c18101e8`.

Table 7: Planned and out-of-scope relations. 28 planned entries (1 declared in the registry with status planned, 27 from the candidate catalog, waves W2 and P); 3 out-of-scope families. Italic names are proposed (the catalog names no entry). None is implemented, so no result in this paper depends on them. Meta properties of relations (catalog section K and 3 rows tagged M, e.g. joint states, costs) are not factors and are excluded; 4 epistemic mechanisms (confidence scaling, soft edges, the reveal / surprise / noise / occlude transforms) are implemented but are not attention factors.